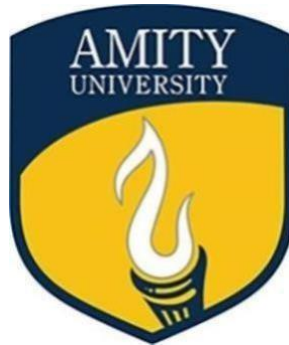


A Project Report

on

Explainable Stroke Risk Prediction In Healthcare Using Machine Learning

Submitted to



Amity University Uttar Pradesh

In partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

Computer Science & Engineering

(Data Science)

by

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DECLARATION

We, “Aneena Sabu, Angel Jain, Anusheka Srivastava ”, students of B.Tech., hereby declare that the project titled “Explainable stroke risk prediction in healthcare using machine learning ” which is submitted by us to Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering(Data Science), has not been previously formed the basis for the reward of any degree, diploma or other similar title or recognition. We hereby declare that we have gone through project guidelines including policy on health and safety, policy on plagiarism, etc.

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ABSTRACT

Stroke is one of the leading causes of death and long-term disability worldwide, making early risk prediction essential for timely medical intervention and improved patient outcomes. Traditional diagnostic approaches often rely on clinical expertise and may not always support rapid identification of high-risk individuals. Recent advancements in Machine Learning (ML) and Explainable Artificial Intelligence (XAI) provide opportunities to develop intelligent systems capable of predicting stroke risk while maintaining transparency in the decision-making process.

This project proposes an explainable stroke risk prediction system using machine learning techniques. The **Healthcare Stroke Prediction Dataset** obtained from Kaggle was used for model development. Before training, the dataset was preprocessed through missing value handling, label encoding, standard scaling, and class balancing using the Synthetic Minority Over-sampling Technique (SMOTE). Six supervised machine learning algorithms—**Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)**—were implemented and evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC score, and Confusion Matrix analysis.

To enhance the transparency and interpretability of the prediction results, **SHapley Additive exPlanations (SHAP)** were integrated into the proposed framework. SHAP provided both global and local explanations by identifying the contribution of individual healthcare features to the prediction outcomes. Experimental evaluation demonstrated that the implemented models achieved varying levels of performance, with **CatBoost** providing the most balanced overall performance, while **Logistic Regression** achieved the highest recall and ROC-AUC score for identifying stroke cases. The explainability analysis further highlighted important stroke risk factors, including age, body mass index, average glucose level, hypertension, heart disease, and smoking status.

The proposed system demonstrates that combining machine learning with explainable artificial intelligence can support accurate, transparent, and reliable stroke risk prediction. The framework has the potential to assist healthcare professionals in early risk assessment and clinical decision-making while providing interpretable predictions that enhance trust in artificial intelligence-based healthcare systems.

Keywords

Stroke Risk Prediction, Machine Learning, Explainable Artificial Intelligence (XAI), SHAP, Healthcare, CatBoost, Logistic Regression, Stroke Prediction, Classification.

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CHAPTER 2. INTRODUCTION

2.1. Background of the study

Stroke is one of the leading causes of death and long-term disability worldwide, and its incidence continues to rise due to increasing cases of hypertension, diabetes, obesity, cardiovascular diseases, and unhealthy lifestyle habits. Despite significant advancements in medical technology and healthcare services, accurately identifying individuals at high risk of stroke before the onset of the disease remains a major challenge. Conventional diagnostic approaches primarily depend on clinical examination, laboratory investigations, medical imaging, and physician expertise. While these methods are effective for diagnosis after symptoms appear, they are often less suitable for early risk prediction and preventive healthcare.

The rapid growth of healthcare data has created opportunities to develop intelligent predictive systems capable of analysing multiple patient attributes simultaneously. Machine learning techniques have demonstrated promising results in predicting stroke risk by learning complex relationships between clinical parameters such as age, hypertension, heart disease, body mass index (BMI), average glucose level, smoking status, and other health-related factors. However, many machine learning models are developed with the primary objective of maximizing prediction accuracy, often overlooking the importance of model interpretability. In healthcare applications, high predictive performance alone is insufficient, as medical professionals require clear explanations for the decisions generated by artificial intelligence systems before they can be trusted in clinical practice.

Another significant challenge in stroke prediction is the imbalance present in healthcare datasets, where the number of stroke cases is considerably lower than the number of non-stroke cases. Such imbalance may cause prediction models to favour the majority class, resulting in poor detection of actual stroke cases. Furthermore, many existing studies focus on a limited number of machine learning algorithms without providing a comprehensive comparison of different models under similar experimental conditions. This makes it difficult to determine which algorithm offers the most appropriate balance between predictive performance and clinical applicability.

Therefore, there is a need to develop an explainable stroke risk prediction framework that not only provides accurate predictions but also offers transparent and interpretable explanations for its decisions. The proposed system addresses this need by implementing and comparing multiple machine learning algorithms, handling class imbalance through appropriate preprocessing techniques, and integrating Explainable Artificial Intelligence (XAI) using SHapley Additive exPlanations (SHAP). This approach aims to support healthcare professionals in making informed decisions while improving trust, transparency, and reliability in AI-assisted stroke risk prediction.

2.2. Introduction to stroke

Stroke is a serious neurological disorder that occurs when the blood supply to a part of the brain is interrupted or significantly reduced, depriving brain tissues of oxygen and essential nutrients. Within a few minutes of this interruption, brain cells begin to die, making stroke a medical emergency that requires immediate diagnosis and treatment. Delayed medical intervention can result in permanent brain damage, physical disability, cognitive impairment, speech difficulties, paralysis, or even death. Due to its severe health consequences and high mortality rate, stroke remains one of the leading causes of long-term disability and death worldwide.

Stroke is broadly classified into two major types: **ischemic stroke** and **hemorrhagic stroke**. Ischemic stroke is the most common type, accounting for nearly 85% of all stroke cases, and occurs when a blood vessel supplying the brain becomes blocked due to a blood clot or fatty deposits. Hemorrhagic stroke, which accounts for the remaining cases, occurs when a weakened blood vessel ruptures, causing bleeding within or around the brain. Both types of stroke require prompt medical attention, although their causes, treatment approaches, and recovery outcomes differ significantly.

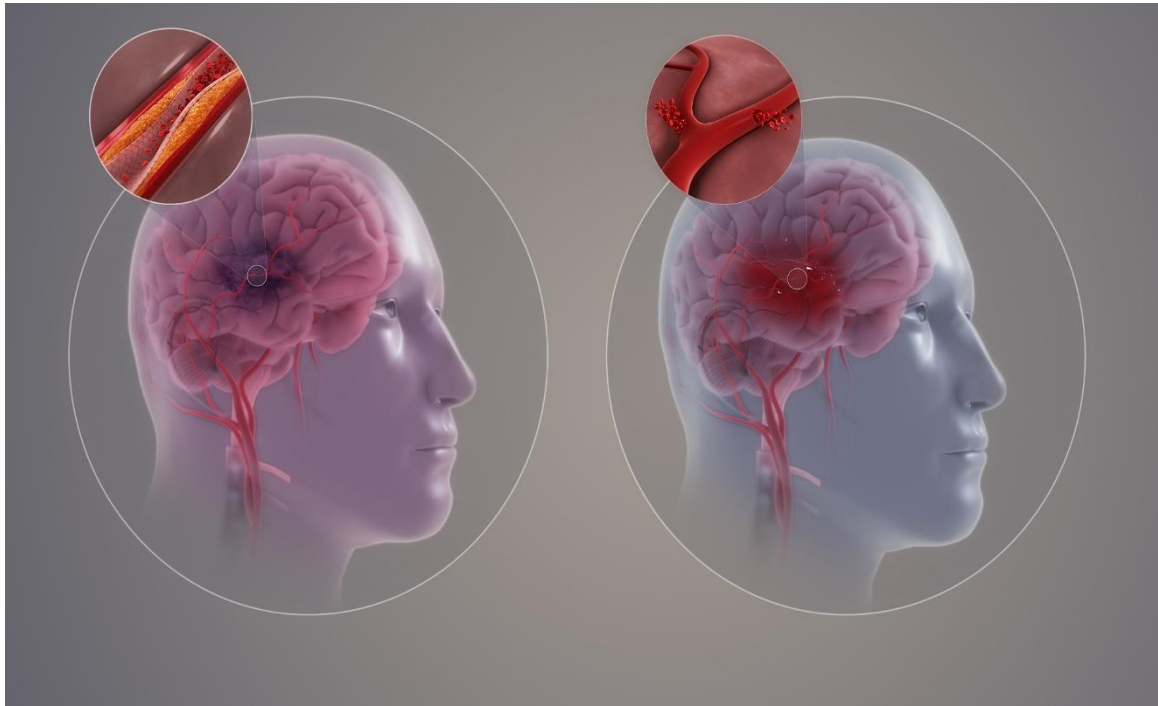


Figure 2.1: Illustration of the two major types of stroke: ischemic stroke and hemorrhagic stroke. Source: from scientific animations via Wikimedia Commons.[23]

Several modifiable and non-modifiable risk factors contribute to the occurrence of stroke. Non-modifiable factors include age, gender, family history, and genetic predisposition, while modifiable factors include hypertension, diabetes, heart disease, obesity, smoking, excessive alcohol consumption, physical inactivity, and elevated blood glucose levels. Among these, hypertension is considered one of the most significant risk factors, as persistently high blood pressure damages blood vessels and increases the likelihood of both ischemic and hemorrhagic strokes. Since many of these risk factors can be managed through lifestyle changes and timely medical intervention, early identification of high-risk individuals plays a crucial role in reducing stroke incidence.

The symptoms of stroke often appear suddenly and vary depending on the affected region of the brain. Common symptoms include sudden weakness or numbness on one side of the body, difficulty in speaking or understanding speech, blurred vision, dizziness, severe headache, loss of balance, and difficulty in walking. Since stroke progresses rapidly, immediate recognition of these warning signs and prompt medical treatment are essential for improving survival rates and reducing long-term complications.

Although advances in medical science have significantly improved stroke diagnosis and

treatment, preventing stroke before its occurrence remains one of the most effective strategies for reducing its overall burden. Early identification of individuals who are at high risk enables healthcare professionals to recommend preventive measures such as lifestyle modifications, continuous health monitoring, and timely medical intervention. Consequently, stroke risk assessment has become an important area of research in modern healthcare, paving the way for the development of intelligent prediction systems capable of supporting preventive clinical decision-making.

2.3. Problem statement

Stroke remains one of the leading causes of mortality and long-term disability worldwide, emphasizing the need for reliable methods for early risk assessment. Although healthcare institutions have access to large volumes of patient data, accurately identifying individuals at high risk of stroke before the occurrence of the disease remains a significant challenge. Traditional risk assessment methods often rely on clinical expertise and manual evaluation of patient records, which can be time-consuming and may not effectively capture the complex relationships among multiple risk factors.

Recent studies have demonstrated the potential of machine learning algorithms for stroke prediction by analysing patient attributes such as age, hypertension, heart disease, body mass index (BMI), average glucose level, and smoking status. However, many existing prediction models primarily focus on improving classification accuracy while providing limited interpretability. In healthcare applications, clinicians require not only accurate predictions but also clear explanations that justify the factors influencing the model's decisions. The lack of transparency in many machine learning models reduces user trust and limits their practical adoption in clinical decision support.

Another challenge is the imbalance present in healthcare datasets, where stroke cases are significantly fewer than non-stroke cases. This imbalance may cause prediction models to favour the majority class, leading to poor identification of actual stroke patients and reducing the effectiveness of the system in real-world healthcare scenarios.

Therefore, there is a need for an explainable stroke risk prediction framework that combines accurate machine learning models with interpretable decision-making

techniques. Such a framework should provide reliable stroke risk prediction while enabling healthcare professionals to understand the factors contributing to each prediction. This study addresses these challenges by comparing multiple machine learning algorithms and integrating Explainable Artificial Intelligence (XAI) through SHAP to improve transparency, reliability, and clinical applicability

2.4. Need for study

The increasing prevalence of stroke and its severe impact on public health highlight the urgent need for effective methods for early risk assessment and prevention. Stroke often occurs without prior warning and can lead to permanent disability or death if not diagnosed and treated promptly. Since many of the associated risk factors, such as hypertension, diabetes, obesity, heart disease, and smoking, can be identified and managed before the occurrence of stroke, early prediction plays a crucial role in reducing its incidence and improving patient outcomes.

Healthcare organizations generate vast amounts of patient data through electronic health records, routine medical examinations, and diagnostic tests. However, analysing this data manually to identify individuals at high risk of stroke can be challenging due to the large number of clinical variables involved. Intelligent machine learning techniques provide an opportunity to analyse these complex datasets efficiently, enabling healthcare professionals to identify high-risk patients more accurately and support preventive decision-making.

In addition to predictive accuracy, transparency has become an essential requirement for the adoption of artificial intelligence in healthcare. Medical professionals need to understand the reasoning behind a model's predictions before incorporating them into clinical practice. Explainable Artificial Intelligence (XAI) addresses this requirement by providing interpretable insights into the factors influencing prediction outcomes, thereby improving trust and supporting informed clinical decisions.

Therefore, this study aims to develop an explainable stroke risk prediction framework that combines multiple machine learning algorithms with SHAP-based explainability. The

proposed system is intended to assist healthcare professionals in the early identification of individuals at risk of stroke while providing transparent and interpretable predictions. Such a system has the potential to support preventive healthcare, improve clinical decision-making, and contribute to better patient management.

2.5. Objective

The primary objective of this study is to develop an explainable machine learning-based framework for early stroke risk prediction using healthcare data. The proposed system aims to accurately identify individuals at risk of stroke while providing transparent explanations of the prediction results through Explainable Artificial Intelligence (XAI) techniques.

The specific objectives of the study are:

- To collect and preprocess a healthcare stroke dataset for machine learning analysis.
- To perform data preprocessing, including handling missing values, encoding categorical variables, feature scaling, and addressing class imbalance using SMOTE.
- To develop stroke prediction models using Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).
- To compare the performance of the developed models using evaluation metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC score.
- To integrate SHAP (SHapley Additive exPlanations) for interpreting model predictions and identifying the most influential features affecting stroke risk.
- To identify the most suitable machine learning model for stroke prediction based on predictive performance and interpretability.
- To propose a deployable framework for an explainable stroke risk prediction system that can support healthcare professionals in clinical decision-making.

2.6. Scope of the study

The scope of this study is limited to the development and evaluation of an explainable machine learning-based framework for predicting stroke risk using a publicly available

healthcare dataset. The proposed system focuses on analysing patient information, including demographic details, medical history, and lifestyle-related factors such as age, gender, hypertension, heart disease, body mass index (BMI), average glucose level, work type, residence type, marital status, and smoking status. These features are used to identify individuals who may be at a higher risk of experiencing a stroke.

The study includes data preprocessing techniques such as handling missing values, encoding categorical variables, feature scaling, and addressing class imbalance using the Synthetic Minority Over-sampling Technique (SMOTE). Multiple supervised machine learning algorithms, namely Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN), are implemented and compared to determine the most suitable model for stroke risk prediction. The performance of these models is evaluated using standard classification metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC score.

To improve transparency and interpretability, the study incorporates Explainable Artificial Intelligence (XAI) using SHapley Additive exPlanations (SHAP). The explainability analysis helps identify the contribution of individual features towards model predictions, thereby making the system more understandable and reliable for healthcare professionals.

Although this study proposes a deployment framework for a web-based stroke prediction system, the implementation is limited to model development, performance evaluation, and deployment planning. The system is not validated using real-time clinical data or integrated with hospital information systems. Furthermore, the proposed framework is intended to support healthcare professionals in decision-making and should not be considered a replacement for medical diagnosis or clinical expertise.

2.7. Motivation

Stroke is a major global health concern due to its high mortality rate and the long-term disabilities it can cause. Despite significant advances in medical science, many stroke cases could be prevented through timely identification of high-risk individuals and appropriate preventive measures. This highlights the importance of developing intelligent systems that

can assist healthcare professionals in identifying patients who may be vulnerable to stroke before the onset of the disease.

The rapid advancement of Machine Learning has demonstrated its potential in analysing complex healthcare datasets and identifying hidden patterns that may not be easily recognized through traditional statistical methods. However, while many machine learning models achieve high prediction accuracy, they often function as **black-box models**, providing little or no explanation for their predictions. In healthcare, where decisions directly affect patient care and treatment, transparency and trust are essential. Healthcare professionals need to understand the reasoning behind a prediction before incorporating it into clinical decision-making.

This study is motivated by the need to bridge the gap between predictive performance and interpretability in stroke risk prediction. By integrating Explainable Artificial Intelligence (XAI) using SHapley Additive exPlanations (SHAP) with multiple machine learning algorithms, the proposed framework aims to provide both accurate predictions and meaningful explanations of the factors influencing those predictions. Such an approach can enhance the confidence of healthcare professionals in AI-assisted systems and support more informed, transparent, and reliable decision-making.

Furthermore, the comparative analysis of multiple machine learning algorithms allows the identification of the most suitable model for stroke prediction based not only on predictive accuracy but also on clinically significant evaluation metrics such as recall, ROC-AUC score, and interpretability. The proposed deployment framework further motivates this research by demonstrating the potential for translating the developed model into a practical decision support system for future healthcare applications.

2.8. Report Organization

This report is organized into nine chapters, each addressing a specific aspect of the proposed explainable stroke risk prediction framework.

Chapter 1 presents the **Abstract**, providing a concise overview of the research problem, objectives, methodology, major findings, and key contributions of the study.

Chapter 2 introduces the research by discussing the background of the study, an overview of stroke, the problem statement, the need for the study, research objectives, scope, motivation, and the overall organization of the report.

Chapter 3 presents the **Literature Review**, covering previous studies on stroke prediction, machine learning applications in healthcare, Explainable Artificial Intelligence (XAI), comparative analysis of selected research papers, and the research gap identified from existing literature.

Chapter 4 describes the **Proposed System and Methodology**, including the dataset, feature description, data preprocessing techniques, machine learning models, SHAP-based explainability, and the performance evaluation metrics used in the study.

Chapter 5 discusses the **System Design and Implementation**, including the system architecture, workflow, experimental setup, software and hardware requirements, model development, and hyperparameter tuning.

Chapter 6 presents the **Experimental Results and Discussion**, including the performance analysis of the implemented machine learning models, confusion matrices, ROC curve analysis, SHAP explainability results, comparative analysis, and interpretation of the experimental findings.

Chapter 7 describes the **Proposed Deployment Framework**, outlining the conceptual design of a deployable stroke risk prediction system, deployment architecture, prediction workflow, potential advantages, implementation challenges, and future enhancements.

Chapter 8 provides the **Conclusion and Future Scope** of the study by summarizing the major findings, contributions, limitations, and possible directions for future research and development.

CHAPTER 3. LITERATURE REVIEW

3.1. Introduction to the Literature

The literature review forms an essential part of this study by providing an understanding of the existing research on stroke risk prediction using Machine Learning (ML) and Explainable Artificial Intelligence (XAI). It helps identify the methodologies, algorithms, preprocessing techniques, evaluation metrics, and explainability approaches adopted by previous researchers, thereby establishing a foundation for the proposed work.

For this study, a comprehensive review of selected conference papers was carried out, focusing on machine learning-based stroke prediction, class imbalance handling techniques, ensemble learning methods, and explainability tools such as SHAP and LIME. The reviewed studies employ various algorithms, including Logistic Regression, Random Forest, Support Vector Machine (SVM), XGBoost, CatBoost, and other advanced learning techniques to improve stroke prediction performance. While many studies have reported high predictive accuracy, several also highlight the importance of model interpretability and the need for transparent AI systems in healthcare.

This chapter presents a brief overview of stroke, discusses the role of machine learning and explainable AI in healthcare, reviews the selected research papers, and identifies the research gap that serves as the basis for the proposed explainable stroke risk prediction framework.

3.2. Stroke prediction research overview

Stroke continues to be one of the most significant public health challenges worldwide due to its high mortality rate and long-term impact on the quality of life of affected individuals. It is a major contributor to permanent disability, often resulting in physical, cognitive, and psychological impairments that require prolonged medical care and rehabilitation. The increasing prevalence of stroke places a substantial burden on healthcare systems, making early risk assessment and prevention an important area of clinical research.

The occurrence of stroke is influenced by multiple demographic, physiological, and lifestyle-related factors. Clinical parameters such as age, hypertension, heart disease, body

mass index (BMI), average glucose level, smoking habits, and other underlying medical conditions collectively contribute to an individual's risk of developing stroke. Since these risk factors interact in complex ways, accurately identifying high-risk individuals using conventional assessment methods can be challenging.

With the availability of large-scale healthcare datasets and advances in computational techniques, researchers have increasingly explored data-driven approaches for stroke risk prediction. Machine learning models are capable of analysing multiple clinical variables simultaneously and identifying patterns that may not be evident through traditional statistical methods. Consequently, stroke prediction has become one of the most actively researched applications of artificial intelligence in healthcare, with the objective of supporting preventive diagnosis and improving clinical decision-making.

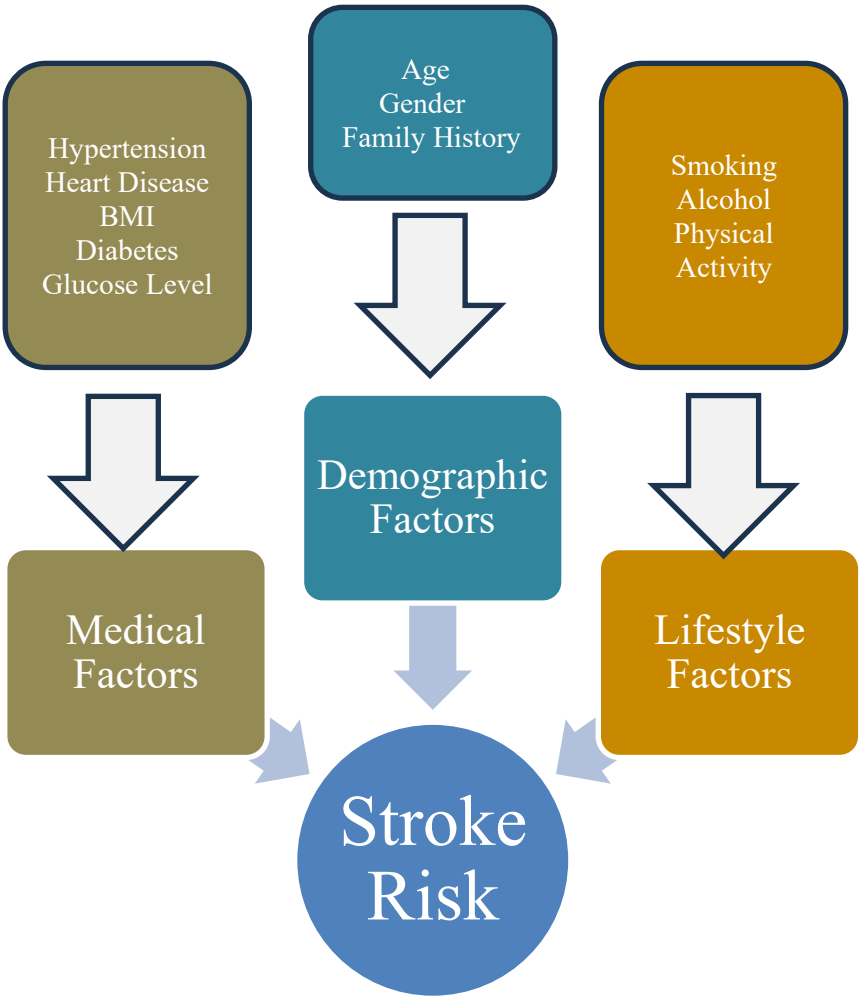


Figure 3.1: Factors influencing stroke risk. Source- Created by the author

3.3. Machine learning approaches for stroke prediction

Machine learning has become one of the most widely adopted approaches for stroke risk prediction due to its ability to analyse healthcare data and identify complex relationships among clinical features. The reviewed studies demonstrate that machine learning techniques improve the accuracy and efficiency of stroke prediction by utilizing demographic information, medical history, physiological measurements, and lifestyle-related factors.

The literature indicates that **supervised machine learning algorithms** are the most commonly used methods for stroke prediction. Hossain *et al.* employed Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) with SMOTE and LIME to improve prediction performance and model interpretability. Similarly, Coffie *et al.* compared machine learning, deep learning, and ensemble models using SHAP and LIME, highlighting the benefits of explainable AI in clinical decision support. Kumar *et al.* proposed the BoostStack framework, combining LightGBM, XGBoost, and CatBoost, and demonstrated improved predictive performance with SHAP-based explanations.

Several other reviewed studies compared algorithms such as Logistic Regression, Random Forest, SVM, XGBoost, and CatBoost. While ensemble models generally achieved higher accuracy, simpler models such as Logistic Regression remained valuable due to their interpretability and consistent performance. The studies also emphasized the importance of preprocessing techniques, including missing value treatment, feature encoding, feature scaling, and SMOTE, to address class imbalance and improve model reliability.

Overall, the reviewed literature confirms that machine learning provides an effective approach for early stroke risk prediction. However, recent studies emphasize that predictive performance alone is insufficient for healthcare applications, highlighting the growing need for explainable and trustworthy models that support clinical decision-making. This has encouraged the integration of Explainable Artificial Intelligence (XAI), discussed in the next section.

Table 3.1: Machine learning algorithms frequently used in the reviewed stroke prediction studies.

Algorithm	Purpose in Reviewed Studies
Logistic Regression	Baseline classification and interpretable prediction
Random Forest	High-accuracy ensemble classification
Support Vector Machine (SVM)	Classification of complex healthcare data
XGBoost	Gradient boosting for improved predictive performance
CatBoost	Efficient handling of categorical healthcare features
K-Nearest Neighbors (KNN)	Distance-based patient classification
Deep Learning Models	Learning complex feature representations
Ensemble Learning	Combining multiple models for better performance

3.4. Explainable AI in stroke prediction

As machine learning models become increasingly sophisticated, ensuring the transparency and interpretability of their predictions has become a major concern in healthcare. The reviewed studies emphasize that while complex models such as ensemble learning and deep learning often achieve high predictive performance, their decision-making process is difficult for clinicians to understand. This limitation has encouraged researchers to integrate Explainable Artificial Intelligence (XAI) techniques into stroke prediction systems.

Among the reviewed papers, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) were the most widely adopted explainability techniques. Hossain *et al.* incorporated LIME to explain individual stroke predictions, while Coffie *et al.* employed both SHAP and LIME to improve the interpretability of machine learning, deep learning, and ensemble models. Similarly, Kumar *et al.* and Dr. V. Shobana *et al.* utilized SHAP to identify the most influential clinical features affecting stroke risk, enabling healthcare professionals to better understand model predictions.

The reviewed literature consistently identifies factors such as age, hypertension, heart disease, body mass index (BMI), average glucose level, and smoking status as the most

significant contributors to stroke prediction. By providing feature-level explanations, XAI techniques increase clinicians' confidence in machine learning models and support more informed clinical decision-making.

Overall, the reviewed studies demonstrate that integrating explainability with machine learning enhances the transparency, reliability, and practical applicability of stroke prediction systems. Consequently, Explainable Artificial Intelligence has become an essential component of modern healthcare decision support systems and forms a key aspect of the proposed research.

3.5. Review of selected papers

This section reviews the selected research papers that form the basis of the proposed study. The reviewed literature focuses on machine learning techniques, ensemble models, data preprocessing, and Explainable Artificial Intelligence (XAI) for stroke risk prediction. These studies collectively demonstrate the growing emphasis on developing accurate, interpretable, and clinically reliable stroke prediction systems.

Hossain et al. (2025)

Hossain *et al.* proposed an automated machine learning framework for early stroke prediction using Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The study integrated SMOTE to address class imbalance and LIME to provide interpretable predictions. Their work demonstrates that combining machine learning with explainable AI can improve the reliability and transparency of stroke risk assessment.

Coffie et al. (2025)

Coffie *et al.* presented a comparative study of machine learning, deep learning, and ensemble learning models for stroke prediction using electronic health records. By incorporating SHAP and LIME, the study emphasized both predictive performance and model interpretability. The research highlighted the growing importance of explainable AI in supporting trustworthy clinical decision-making.

Kumar et al. (2025)

Kumar *et al.* introduced the BoostStack framework, which combines LightGBM, XGBoost, and CatBoost to improve stroke prediction. The proposed ensemble model employed SHAP to explain prediction outcomes, demonstrating that hybrid learning approaches can enhance both prediction accuracy and interpretability in healthcare applications.

Shobana et al. (2025)

Shobana *et al.* investigated the integration of Explainable Artificial Intelligence in stroke diagnosis and prevention using multiple machine learning algorithms. The study demonstrated how explainability techniques such as SHAP, LIME, and ELI5 improve transparency, enabling clinicians to better understand model predictions and make informed healthcare decisions.

Prentzas et al. (2019)

Prentzas *et al.* proposed an explainable stroke prediction model by integrating machine learning with symbolic reasoning. Unlike conventional black-box models, the study generated rule-based explanations that enhanced transparency while maintaining reliable prediction performance. This work highlighted the potential of combining predictive intelligence with human-understandable reasoning in healthcare.

Swarnima et al. (2025)

Swarnima *et al.* focused on reducing bias in stroke prediction models through Explainable Artificial Intelligence techniques. Using SHAP and LIME, the study identified biased features influencing model predictions and demonstrated that explainability methods contribute to fairer, more reliable, and clinically acceptable stroke prediction systems.

Mow et al. (2025)

Mow *et al.* developed an interpretable stroke prediction model using SMOTEENN to address class imbalance and advanced machine learning algorithms for classification. By

integrating both SHAP and LIME, the proposed approach improved prediction reliability while providing meaningful explanations that support clinical interpretation of stroke risk.

Advithi et al. (2024)

Advithi *et al.* investigated the combined use of SMOTE and LIME for improving stroke prediction performance. The study demonstrated that balancing the dataset significantly enhanced model performance, while LIME increased the transparency of machine learning predictions by explaining individual classification outcomes.

Zamil et al. (2025)

Zamil *et al.* proposed an interpretable machine learning framework for stroke prediction using Logistic Regression, Random Forest, XGBoost, and CatBoost. The study incorporated SMOTE for handling class imbalance and SHAP for explaining feature importance. The findings demonstrated that combining balanced datasets with explainable AI improves both prediction performance and model transparency.

Purkayastha et al. (2025)

Purkayastha *et al.* developed a comprehensive stroke prediction framework integrating ensemble learning, SHAP-based interpretability, and bias assessment. The study emphasized that accurate prediction models should also ensure fairness and transparency, highlighting the importance of explainable AI in developing trustworthy healthcare decision support systems.

K. M. Mahabub Hossain et al. (2025)

K. M. Mahabub Hossain *et al.* proposed a multiclass stroke risk prediction model using LightGBM, XGBoost, Support Vector Machine, and ensemble learning techniques. The framework incorporated SHAP for advanced interpretability, enabling clinicians to identify the clinical features that most strongly influence stroke risk predictions.

Bhagya et al. (2026)

Bhagya *et al.* introduced the AI-XAI-LLM framework for stroke prediction by integrating machine learning, SHAP-based explanations, and a Large Language Model (LLM). The proposed approach translated technical explainability outputs into natural language, making prediction results easier for clinicians to understand and improving trust in AI-assisted healthcare systems.

Arun et al. (2024)

Arun *et al.* compared multiple machine learning algorithms for stroke prediction while employing SHAP and LIME to improve model interpretability. The study demonstrated that combining explainability techniques with high-performing classifiers enhances transparency and supports clinicians in understanding prediction outcomes.

Islam et al. (2023)

Islam *et al.* focused on improving stroke prediction by integrating Random Forest with Explainable Artificial Intelligence techniques. Using SHAP and LIME, the study provided interpretable visual explanations that helped distinguish stroke and non-stroke cases while enhancing the reliability of machine learning predictions

Rathore et al. (2025)

Rathore *et al.* developed an interpretable stroke prediction model using boosting algorithms integrated with Explainable Artificial Intelligence techniques. The framework employed SHAP, LIME, ALE, and ELI5 to explain prediction outcomes, demonstrating that boosting methods can achieve high predictive performance while maintaining model transparency.

Ugbomeh et al. (2024)

Ugbomeh *et al.* investigated multiple machine learning algorithms for stroke risk prediction using Explainable Artificial Intelligence techniques. The study combined SMOTE with LIME and ELI5 to improve prediction reliability and interpretability. The findings reinforced the significance of explainable machine learning in identifying critical stroke risk factors and supporting clinical decision-making

3.6. Comparative literature table

Table 3.2: Comparative analysis of the research papers on stroke risk prediction.

Paper Author	Paper Title	Publication Year	Objective	Methodology	Outcome
Saber Hossain et al	Automated Stroke Prediction Using Machine Learning Approach	2025	To develop an automated machine learning-based system for early stroke prediction using healthcare data and explainable AI techniques.	Applied Random Forest, SVM, and KNN algorithms on a preprocessed stroke dataset(kaggle sourced) using SMOTE for class imbalance handling and LIME for model interpretability. Hyperparameter tuning and performance evaluation were also performed.	Random Forest achieved the best performance with 97% accuracy and improved interpretability using LIME-based explanations for stroke risk prediction.
Lord Coffie et al	Comparative Analysis of Machine Learning, Deep Learning, and Ensemble Models for Stroke Prediction Using Electronic Health Records	2025	To compare ML, DL, and ensemble models for accurate and interpretable stroke prediction using structured healthcare data.	Applied ML models (RF, SVM, XGBoost, LR), ANN-based deep learning models, and a stacked ensemble model with SHAP and LIME interpretability techniques using EHR data and SMOTE preprocessing	The stacked ensemble model achieved the best performance, while SHAP and LIME identified age, glucose level, hypertension, and BMI as major stroke risk factors.
Davinder Kumar et al	Enhanced Hybrid BoostStack Framework for Accurate and Early Prediction of Brain Stroke Risk	2025	To develop a hybrid ensemble framework for accurate and early prediction of brain stroke risk using machine learning techniques.	Implemented a BoostStack framework integrating LightGBM, XGBoost, CatBoost, and stacking ensemble learning with SMOTE preprocessing, feature selection, and SHAP-based interpretability.	The proposed framework achieved 97.2% accuracy, 95.8% precision, 96.5% recall, and improved interpretability for real-time stroke risk assessment.
Dr V Shobana et al	"Integrating Explainable Machine Learning (XAI) in Stroke Medicine: Opportunities and Challenges for Early Diagnosis and Prevention"	2025	To explore the role of explainable AI techniques in improving early stroke diagnosis, prevention, and clinical decision-making.	Applied traditional ML models such as KNN, Decision Tree, Random Forest, SVM, and Logistic Regression with preprocessing, SMOTE balancing, and XAI techniques including SHAP, LIME, and ELI5 on a healthcare stroke dataset.	Logistic Regression achieved better classification performance, while XAI techniques improved transparency and identified age, BMI, and glucose level as major stroke risk factors for reliable clinical interpretation.
Nicoletta Prentzas et al	Integrating ml with symbolic reasoning to build an explainable AI model for stroke prediction	2019	To develop an explainable AI model for stroke prediction by integrating machine learning with symbolic reasoning techniques.	Applied Random Forest for stroke prediction and extracted decision rules using the inTrees framework. Integrated symbolic reasoning and Gorgias-based argumentation theory to generate interpretable explanations for predictions.	The proposed explainable argumentation system achieved comparable prediction performance while providing transparent rule-based explanations for stroke risk assessment and clinical decision-making.
Swarnima et al	"Mitigating Bias in Brain Stroke Prediction Using Explainable Artificial Intelligence"	2025	To identify and reduce bias in stroke prediction models using explainable AI techniques for more reliable clinical decision-making.	Applied Logistic Regression, Naïve Bayes, and Random Forest models on balanced and imbalanced stroke datasets using SHAP and LIME for explainability and oversampling techniques to reduce dataset bias	Explainable AI helped identify biased features and improve model performance. The final Random Forest model achieved an F1-score of 85.01% with improved prediction reliability and interpretability.
Momtaz Hossain Mow et al	"An Interpretable Stroke Risk Prediction Model using SMOTEENN and Advanced ML: An XAI-Driven Approach"	2025	To develop an accurate and interpretable stroke risk prediction model using machine learning while addressing class imbalance and improving clinical transparency.	Preprocessed Kaggle stroke dataset (handling missing values, encoding, scaling, outlier removal), applied SMOTEENN for class imbalance, trained ML models (XGBoost, Random Forest, CatBoost) with hyperparameter tuning, and used LIME & SHAP for explainability.	XGBoost achieved the best performance (96% accuracy, AUC 0.94), with improved prediction reliability and interpretability, supporting better clinical decision-making.
Advithi D	"Integrating SMOTE and	2024	To improve stroke prediction by	Preprocessed dataset (handling missing	Random Forest performed

et al	LIME Techniques for Enhanced Stroke Prediction using Machine Learning Approaches"		addressing class imbalance using SMOTE and enhancing model interpretability with LIME.	BMI, encoding, Z-scaling), applied SMOTE for balancing, trained multiple ML models (LR, SVM, RF, XGBoost, etc.), conducted 3 experiments (with/without scaling & SMOTE), and used LIME for explainability.	best (94.7% accuracy with SMOTE + scaling); SMOTE improved model performance overall, and LIME provided interpretable feature insights for predictions.
Zakir Hossain Zamil et al	"Stroke Prediction on Healthcare Data Using SMOTE and Explainable Machine Learning"	2025	To develop an efficient and interpretable machine learning system for early stroke risk prediction using healthcare data.	Applied Logistic Regression, Random Forest, XGBoost, and CatBoost models on a SMOTE-balanced healthcare dataset with SHAP-based explainability and feature importance analysis.	CatBoost achieved the best performance with 95.58% accuracy and 95.51% F1-score, while SHAP identified age, glucose level, hypertension, heart disease, and BMI as important stroke risk factors.
Debolina Purkayastha et al	"A Comprehensive Approach to Stroke Prediction using Machine Learning: Integrating Ensemble Learning, SHAP interpretability and Bias Assessment"	2025	To develop an accurate, interpretable, and fair stroke prediction system using ensemble machine learning and explainable AI techniques.	Applied Logistic Regression, Random Forest, and XGBoost models with SMOTE-based preprocessing, SHAP interpretability, and fairness/bias assessment on a stroke healthcare dataset.	Random Forest achieved the highest accuracy (97.38%), while XGBoost showed better fairness performance across subgroups. SHAP identified important stroke risk factors and improved model interpretability.
K M Mahabub Hossain et al	A Machine Learning and Explainable AI-Based Stroke Risk Prediction Approach With SHAP Based Advanced Interpretability	2025	To develop a multiclass stroke risk prediction system using machine learning and explainable AI techniques for improved prediction accuracy and interpretability.	Applied LightGBM, XGBoost, SVM, and ensemble models with SMOTE-NC, Random Forest-based feature selection, Bayesian hyperparameter tuning, and SHAP-based interpretability on a multiclass healthcare stroke dataset.	LightGBM achieved the best performance with 93.94% accuracy and 93.90% F1-score, while SHAP improved interpretability by identifying important stroke risk factors
Thilini Bhagya et al	AI-XAI-LLM: Interpretable Insights into Stroke Risk Prediction	2026	To develop an AI-XAI-LLM framework for accurate stroke risk prediction with clinician-friendly explanations.	Used Kaggle Stroke Prediction Dataset, Random Forest model, SHAP for explainability, and Gemma LLM to convert SHAP outputs into natural language explanations.	Achieved high prediction accuracy (95.76%) and improved clarity and trust of explanations for clinicians compared to raw SHAP outputs.
Chalana B Arun et al	"Comparison of Various Computational Models for Accurate Stroke Prediction and Interpretation using LIME and SHAP Kernels" To improve stroke prediction accuracy and interpretability using Random Forest and Explainable AI techniques like SHAP and LIME	2024	To compare multiple machine learning models for accurate stroke prediction and improve interpretability using LIME and SHAP techniques.	Used Kaggle Healthcare Stroke Dataset with preprocessing, EDA, and ML models such as Random Forest, KNN, SVM, Naïve Bayes, AdaBoost, XGBoost, etc. LIME and SHAP were applied for explainable AI interpretation.	Random Forest and AdaBoost achieved the best performance, with Random Forest reaching 95.64% accuracy after hyperparameter tuning. LIME and SHAP improved model transparency and interpretability.
Nazmin Islam et al	Enhancing Stroke Prediction through Interpretable AI: Distinguishing Stroke Cases from Non-Stroke Cases	2023	To improve stroke prediction accuracy and interpretability using Random Forest and Explainable AI techniques like SHAP and LIME	Used Kaggle Stroke Prediction Dataset with preprocessing, Random Oversampling for class imbalance, five-fold cross-validation, Random Forest classifier, and XAI methods (SHAP & LIME).	Achieved 94.65% accuracy with high recall and improved transparency in predictions through explainable AI visualizations and interpretations.
Shiwangi Rathore et al	Integrating Explainable AI with Ensemble Boosting Models for Early Detection of Brain Stroke	2025	To develop an interpretable boosting-based machine learning model for early stroke risk prediction using Explainable AI techniques.	Used Kaggle Stroke Dataset with preprocessing, class balancing, boosting algorithms (GB, XGB, AdaBoost, HB), and XAI methods including SHAP, LIME, ALE, and ELI5.	XGBoost achieved the best performance with high accuracy and stability, while SHAP, LIME, ALE, and ELI5 improved transparency and interpretability of stroke predictions.
Ogochukwa Ugboemeh et al	Machine Learning Algorithms for Stroke Risk Prediction Leveraging on Explainable Artificial Intelligence Techniques	2024	To predict stroke risk using machine learning models and improve interpretability through Explainable AI (XAI) techniques.	Used Kaggle Stroke Dataset with SMOTE for class balancing, applied six ML models including Random Forest and XGBoost, and used LIME and ELI5 for explanation of predictions.	Random Forest achieved the best performance with 94.5% accuracy and XAI techniques identified key stroke risk factors such as

	(XAI)				age, glucose level, BMI, marital status
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3.7. Research Gap

The literature review indicates that significant progress has been made in applying machine learning and Explainable Artificial Intelligence (XAI) for stroke risk prediction. Many researchers have achieved high prediction accuracy by employing advanced machine learning algorithms, ensemble learning techniques, and data preprocessing methods such as SMOTE. Furthermore, explainability techniques including SHAP, LIME, and ELI5 have improved the transparency of prediction models, enabling clinicians to better understand model decisions.

Despite these advancements, several research gaps remain. Most studies focus primarily on maximizing predictive accuracy using a single algorithm or a specific ensemble framework, while limited attention is given to conducting a comprehensive comparison of multiple machine learning algorithms under the same preprocessing conditions. Such comparisons are essential for identifying the most suitable model based not only on accuracy but also on evaluation metrics such as precision, recall, F1-score, and ROC-AUC, which are particularly important in healthcare applications.

Another limitation observed in the reviewed literature is that many studies evaluate explainability using a single XAI technique. Although SHAP and LIME are widely used, few studies systematically compare model performance while integrating explainability into the complete prediction framework. In addition, most existing research focuses on model development and evaluation, with limited discussion on how these models can be translated into practical, user-friendly systems for healthcare professionals.

To address these gaps, the proposed study develops an explainable stroke risk prediction framework that compares six machine learning algorithms—Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—using a common preprocessing pipeline. The framework integrates SHAP-based explainability to enhance model transparency and proposes a deployment framework that

can support future clinical decision-making and real-world implementation.

CHAPTER 4. PROPOSED SYSTEM AND METHODOLOGY

4.1. Overview

This chapter presents the methodology adopted for developing an explainable stroke risk prediction system using machine learning techniques. The proposed methodology includes dataset acquisition, data preprocessing, model development, explainability analysis, and performance evaluation. Each stage is designed to improve prediction accuracy while ensuring transparency and interpretability for healthcare applications.

The **Healthcare Stroke Prediction Dataset** is used in this study. Before training the models, the dataset is preprocessed by handling missing values, encoding categorical variables, standardizing numerical features, and balancing the data using the Synthetic Minority Over-sampling Technique (SMOTE). These preprocessing steps enhance data quality and improve model performance.

Six supervised machine learning algorithms—**Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)**—are implemented and evaluated using performance metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC score, and Confusion Matrix. To improve model interpretability, **SHapley Additive exPlanations (SHAP)** is employed to explain the contribution of individual features to stroke risk prediction.

4.2. Dataset

The proposed stroke risk prediction system utilizes the **Healthcare Stroke Prediction Dataset**, which was obtained from the Kaggle platform. The dataset is widely used for machine learning research because it contains a combination of demographic, medical, and lifestyle-related factors that influence an individual's risk of stroke. It is designed as a binary classification dataset, where the objective is to predict whether a patient is likely to experience a stroke based on the available healthcare attributes.

The dataset consists of **5,110 patient records** with **11 input features** and **one target**

variable. The input features include demographic information such as age and gender, medical conditions such as hypertension and heart disease, and lifestyle-related factors including smoking status, body mass index (BMI), average glucose level, work type, marital status, and residence type. The target variable, **stroke**, is represented as **0** (No Stroke) and **1** (Stroke).

Since the dataset contains both numerical and categorical variables, appropriate preprocessing techniques were applied before model development. The dataset was examined for missing values, categorical attributes, and class imbalance to ensure that it was suitable for machine learning analysis. The preprocessing steps adopted in this study are discussed in the subsequent sections of this chapter.

Table 4.1 Dataset Summary

Parameter	Description
Dataset Name	Healthcare Stroke Prediction Dataset
Source	Kaggle
Total Records	5,110
Number of Features	11
Target Variable	Stroke
Target Classes	0 – No Stroke, 1 – Stroke
Dataset Type	Binary Classification
Data Types	Numerical and Categorical

4.3. Feature description

The Healthcare Stroke Prediction Dataset consists of **11 input features** and **one target variable**. These features represent demographic characteristics, medical history, and lifestyle-related factors that may influence an individual's risk of stroke. A brief description of each feature is provided in **Table 4.2**.

Table 4.2 Feature Description of the Healthcare Stroke Prediction Dataset

Feature	Description	Data Type
gender	Gender of the patient (Male, Female, Other)	Categorical
age	Age of the patient in years	Numerical
hypertension	Indicates whether the patient has hypertension (0 = No, 1 = Yes)	Binary
heart_disease	Indicates whether the patient has heart disease (0 = No, 1 = Yes)	Binary
ever_married	Marital status of the patient (Yes/No)	Categorical
work_type	Type of employment (Private, Self-employed, Govt_job, Children, Never_worked)	Categorical
Residence_type	Type of residence (Urban/Rural)	Categorical
avg_glucose_level	Average glucose level measured in the blood	Numerical
bmi	Body Mass Index of the patient	Numerical
smoking_status	Smoking habit (Never smoked, Formerly smoked, Smokes, Unknown)	Categorical
stroke	Target variable indicating stroke occurrence (0 = No Stroke, 1 = Stroke)	Binary

The selected features were chosen because they represent important demographic, medical, and lifestyle factors associated with stroke risk. These variables were used as input attributes for training the machine learning models, while the **stroke** attribute served as the target variable for classification. Before model development, the dataset underwent preprocessing to handle missing values, encode categorical variables, standardize numerical features, and address class imbalance.

4.4. Data collection

The dataset used in this study was obtained from the **Healthcare Stroke Prediction Dataset** available on the Kaggle platform. The dataset was selected because it contains a comprehensive set of demographic, medical, and lifestyle-related attributes commonly associated with stroke risk. It is publicly available and widely used for research on machine learning-based healthcare prediction.

After downloading the dataset, it was imported into the Python environment using the

Pandas library for further analysis. An initial examination of the dataset was performed to understand its structure, identify missing values, verify data types, and assess class distribution. The collected data was then prepared for preprocessing before being used to train and evaluate the machine learning models.

Source: Healthcare Stroke Prediction Dataset, Kaggle

4.5. Data Preprocessing

Data preprocessing is an essential step in the machine learning pipeline, as the quality of the input data directly influences model performance. Raw datasets often contain inconsistencies such as missing values, categorical variables, and imbalanced class distributions, which can negatively affect the accuracy and reliability of predictive models. Therefore, the Healthcare Stroke Prediction Dataset was preprocessed before training the machine learning models.

The preprocessing procedure adopted in this study involved handling missing values, encoding categorical variables into numerical representations, standardizing numerical features, and balancing the dataset using the Synthetic Minority Over-sampling Technique (SMOTE). These steps improved the quality of the dataset and ensured that it was suitable for machine learning model development. The individual preprocessing techniques employed in this study are discussed in the following sections.

4.6. Missing values

Missing values are a common issue in healthcare datasets and can affect the performance of machine learning models if not handled appropriately. Therefore, the Healthcare Stroke Prediction Dataset was examined to identify attributes containing missing data before model development.

During the analysis, it was observed that the Body Mass Index (BMI) attribute contained missing values, while the remaining features were complete. To preserve the integrity of the dataset and avoid unnecessary loss of information, the missing BMI values were replaced using the median value of the corresponding feature. The median was selected

because it is less sensitive to extreme values and provides a robust measure of central tendency for numerical data.

After handling the missing values, the dataset was verified to ensure that no null values remained. This preprocessing step improved the completeness and quality of the dataset, making it suitable for subsequent preprocessing and machine learning model training.

Table 4.3: Missing Values in the Dataset

Feature	Missing Values	Treatment Applied
BMI	201	Replaced using Median
Remaining Features	0	No treatment required

4.7. Label encoding

Machine learning algorithms require numerical input data for model training and prediction. Since the Healthcare Stroke Prediction Dataset contains several categorical attributes, these variables were converted into numerical form using the **Label Encoding** technique.

The categorical features encoded in this study include **gender**, **ever_married**, **work_type**, **Residence_type**, and **smoking_status**. Each unique category was assigned a corresponding numerical value while preserving the information contained within the original feature. This transformation enabled the machine learning algorithms to process the data efficiently without altering the underlying relationships among the categories.

Label encoding simplified the dataset and ensured compatibility with all the implemented machine learning models. After encoding, all input features were represented in numerical format, making the dataset ready for the subsequent preprocessing steps.

Table 4.4 Encoded Categorical Features

Feature	Original Values
Gender	Male, Female, Other

Feature	Original Values
Ever Married	Yes, No
Work Type	Private, Self-employed, Govt_job, Children, Never_worked
Residence Type	Urban, Rural
Smoking stat	Never Smoked, Formerly Smoked, Smokes, Unknown

4.8. Standard scaling

Standard Scaling is a feature scaling technique used to normalize numerical attributes so that they have a mean of zero and a standard deviation of one. Since the features in the Healthcare Stroke Prediction Dataset have different ranges and units, scaling was performed to prevent variables with larger values from dominating the learning process of the machine learning models.

In this study, **StandardScaler** from the Scikit-learn library was applied to the numerical features, including **age**, **average glucose level**, and **BMI**. Standardizing these features improved data consistency and enabled models such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Logistic Regression to perform more effectively.

After standard scaling, the numerical features were transformed into a standardized format while preserving the relationships between the data points. The scaled dataset was then used for training and evaluating the machine learning models.

Table 4.5 Standard Scaled Features

Feature	Reason for Scaling
Age	Large range of values
Average Glucose Level	High numerical variation
BMI	Different measurement scale

4.9. SMOTE

The Healthcare Stroke Prediction Dataset exhibited a significant class imbalance, with the majority of records belonging to the **non-stroke** class and only a small proportion

representing **stroke** cases. Such an imbalance can cause machine learning models to become biased toward the majority class, resulting in poor detection of actual stroke cases.

To overcome this issue, the **Synthetic Minority Over-sampling Technique (SMOTE)** was applied to the training dataset. SMOTE generates synthetic samples of the minority class by interpolating between existing minority class instances instead of simply duplicating them. This process increases the representation of the minority class and helps create a more balanced dataset for model training.

Applying SMOTE improved the ability of the machine learning models to learn the characteristics of stroke cases, thereby enhancing their capability to correctly identify patients at risk. The balanced dataset contributed to improved classification performance, particularly in terms of **recall** and **ROC-AUC**, which are essential evaluation metrics for healthcare prediction systems.

4.10. Machine learning models

Machine learning models play a significant role in predicting stroke risk by identifying patterns and relationships within healthcare data. In this study, six supervised machine learning algorithms were implemented to classify patients into stroke and non-stroke categories. These algorithms were selected because they employ different learning mechanisms and provide a comprehensive comparison of classification performance. Each model was trained using the preprocessed dataset and later evaluated using multiple performance metrics. A brief description of each algorithm is provided below.

4.10.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm widely used for binary classification problems. It estimates the probability of an outcome using the sigmoid function, producing values between 0 and 1. Due to its simplicity, computational efficiency, and interpretability, Logistic Regression is commonly applied in medical diagnosis and healthcare prediction tasks.

4.10.2 Random Forest

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions using majority voting. The algorithm reduces overfitting and improves classification stability by utilizing bootstrap sampling and random feature selection. It is widely used for healthcare prediction because of its ability to model complex relationships within data.

4.10.3 XGBoost

Extreme Gradient Boosting (XGBoost) is an advanced ensemble learning algorithm based on gradient boosting. It builds decision trees sequentially, where each tree attempts to minimize the errors made by previous trees. XGBoost incorporates regularization techniques to improve model generalization and computational efficiency, making it suitable for large-scale classification problems.

4.10.4 CatBoost

CatBoost is a gradient boosting algorithm developed to efficiently handle categorical features while reducing prediction bias and overfitting. It employs ordered boosting and effective encoding techniques to improve learning performance. CatBoost has been widely adopted in classification tasks due to its robustness and ability to process mixed-type datasets.

4.10.5 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm that classifies data by determining the optimal hyperplane that maximizes the separation between different classes. The algorithm is effective for both linear and non-linear classification problems through the use of kernel functions and is widely applied in medical data analysis.

4.10.6 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a non-parametric supervised learning algorithm that classifies a data instance based on the majority class of its nearest neighboring data points.

The algorithm relies on distance-based similarity and is simple to implement, making it a commonly used baseline classifier for comparative studies.

4.11. SHAP

SHapley Additive exPlanations (SHAP) is an Explainable Artificial Intelligence (XAI) technique used to interpret the predictions made by machine learning models. It is based on Shapley values from cooperative game theory, which assign an importance value to each feature according to its contribution to the model's prediction. SHAP provides both global and local explanations, enabling a better understanding of how individual features influence the prediction outcomes.

In this study, SHAP was employed to improve the transparency and interpretability of the developed machine learning models. By analysing feature contributions, SHAP helps identify the most influential healthcare parameters affecting stroke prediction, allowing the decision-making process of the models to be understood more clearly. This is particularly important in healthcare applications, where clinicians require interpretable and trustworthy predictions before making medical decisions.

To analyse model behaviour, SHAP visualizations such as the **Summary Plot**, **Bar Plot**, and **Waterfall Plot** were generated. These visualizations provide insights into overall feature importance as well as the contribution of individual features to specific predictions, thereby enhancing confidence in the developed stroke risk prediction system.

4.12. Performance metrics

The performance of the machine learning models was evaluated using several classification metrics to determine their effectiveness in predicting stroke risk. Since stroke prediction is a binary classification problem, relying on a single evaluation metric such as accuracy may not provide a complete assessment of model performance. Therefore, multiple performance metrics were considered to evaluate the models comprehensively.

4.12.1 Accuracy

Accuracy measures the proportion of correctly classified instances among the total number of observations. It provides an overall indication of model performance but may not be sufficient for imbalanced datasets such as stroke prediction.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

4.12.2 Precision

Precision measures the proportion of correctly predicted positive cases out of all instances predicted as positive. A higher precision indicates fewer false positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

4.12.3 Recall (Sensitivity)

Recall measures the proportion of actual positive cases that are correctly identified by the model. In healthcare applications, recall is particularly important because failing to detect patients at risk of stroke may have serious consequences.

$$\text{Recall} = \frac{TP}{TP + FN}$$

4.12.4 F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of model performance, especially when dealing with imbalanced datasets.

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

4.12.5 ROC-AUC Score

The Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) measures the ability of a model to distinguish between positive and negative classes across different classification thresholds. A higher ROC-AUC score indicates better classification performance.

4.12.6 Confusion Matrix

A Confusion Matrix is used to evaluate the classification performance of a model by comparing predicted class labels with the actual class labels. It consists of four components:

- True Positive (TP): Stroke cases correctly predicted as stroke.
- True Negative (TN): Non-stroke cases correctly predicted as non-stroke.
- False Positive (FP): Non-stroke cases incorrectly predicted as stroke.
- False Negative (FN): Stroke cases incorrectly predicted as non-stroke.

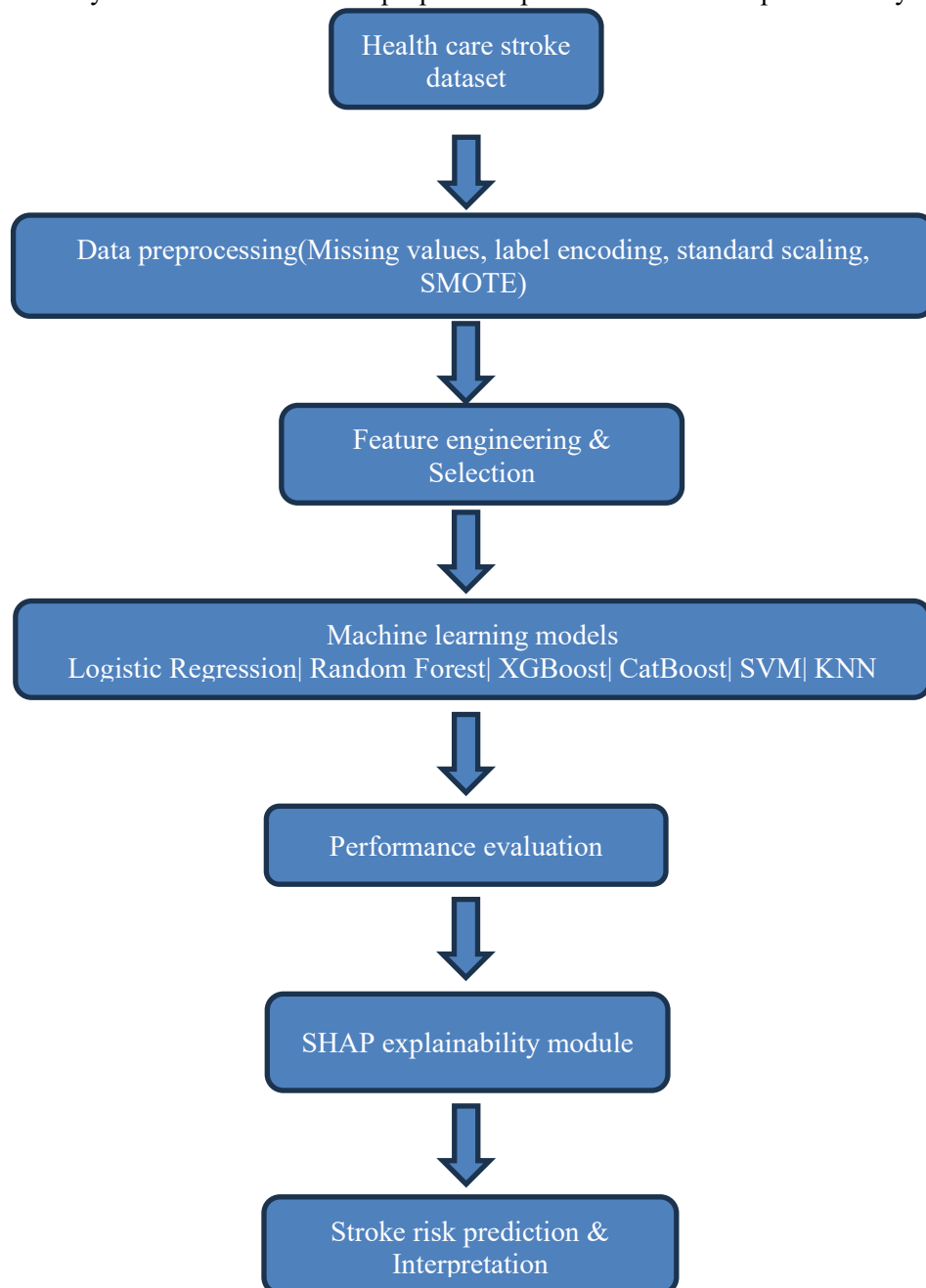
These evaluation metrics provide a comprehensive assessment of the machine learning models and facilitate a fair comparison of their performance in predicting stroke risk.

CHAPTER 5. SYSTEM DESIGN AND IMPLEMENTATION

5.1. System Architecture

The system architecture illustrates the overall structure of the proposed explainable stroke risk prediction system. It represents the interaction between the dataset, preprocessing techniques, machine learning models, explainability module, and prediction output. The architecture demonstrates how patient data is processed through different stages to generate an interpretable stroke risk prediction

Figure 5.1: System architecture of the proposed explainable stroke risk prediction system.



5.2. Workflow

The workflow of the proposed system consists of a sequence of stages beginning with dataset acquisition and ending with stroke risk prediction. Initially, the healthcare stroke dataset is collected and preprocessed to improve data quality. The preprocessing stage includes handling missing values, encoding categorical variables, standardizing numerical features, and balancing the dataset using SMOTE.

The processed dataset is then used to train six supervised machine learning models: Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). The trained models are evaluated using multiple performance metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC score, and Confusion Matrix. Finally, SHAP is applied to explain the contribution of each feature towards the prediction results, enabling transparent and interpretable stroke risk prediction.

5.3. Experimental setup

The experimental work for this study was carried out using the **Kaggle Notebook** environment with the Python programming language. Kaggle provides a cloud-based platform that supports data analysis and machine learning, enabling model development without requiring local software installation. The Healthcare Stroke Prediction Dataset was imported into the Kaggle environment, where preprocessing, model training, and performance evaluation were performed.

Several Python libraries were used during the implementation. **Pandas** and **NumPy** were employed for data manipulation and numerical computations, while **Matplotlib** and **Seaborn** were used for data visualization. Machine learning models were implemented using the **Scikit-learn** library, with **XGBoost** and **CatBoost** libraries used for their respective algorithms. The **SHAP** library was integrated to interpret model predictions and identify the contribution of individual features towards stroke risk prediction.

The dataset was preprocessed by handling missing values, encoding categorical variables, standardizing numerical features, and balancing the training data using SMOTE. Six supervised machine learning models—**Logistic Regression, Random Forest, XGBoost,**

CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—were trained and evaluated using multiple performance metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC score, and Confusion Matrix.

Table 5.1 Experimental Environment

Component	Specification
Programming Language	Python
Development Environment	Kaggle
Dataset	Healthcare Stroke Prediction Dataset
Libraries Used	Pandas, NumPy, Scikit-learn, XGBoost, CatBoost, SHAP, Matplotlib, Seaborn
Machine Learning Models	Logistic Regression, Random Forest, XGBoost, CatBoost, SVM, KNN
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix

5.4. Software requirement

The implementation of the proposed stroke risk prediction system required various software tools and libraries for data preprocessing, machine learning model development, visualization, and explainability analysis.

Table 5.2 Software Requirements

Software/Tool	Purpose
Operating System	Windows 10
Development Platform	Kaggle Notebook
Programming Language	Python
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Scikit-learn	Machine learning model implementation
XGBoost	XGBoost classifier implementation
CatBoost	CatBoost classifier implementation

Software/Tool	Purpose
SHAP	Model explainability and feature interpretation
Matplotlib	Data visualization
Seaborn	Statistical data visualization

5.5. Hardware requirements

The proposed stroke risk prediction system was implemented using a personal computer with internet connectivity to access the Kaggle Notebook environment. Since the experiments were performed on a cloud-based platform, no specialized hardware was required. The minimum hardware requirements are listed in Table 5.3.

Table 5.3 Hardware Requirements

Components	Purpose
Platform	Kaggle Notebook
Processor	Intel core i5 or equivalent
RAM	8GB
Operating System	Windows 10

5.6. Model development

The machine learning models were developed using the preprocessed Healthcare Stroke Prediction Dataset. After completing data preprocessing, the dataset was divided into training and testing sets to enable model training and performance evaluation. Six supervised machine learning algorithms—Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—were implemented using the Scikit-learn, XGBoost, and CatBoost libraries in Python.

Each model was trained using the same training dataset to ensure a fair comparison of their predictive capabilities. After training, the models were evaluated on the testing dataset using multiple performance metrics, including Accuracy, Precision, Recall, F1-score,

ROC-AUC score, and Confusion Matrix. In addition, SHapley Additive exPlanations (SHAP) was employed to interpret the prediction outcomes and identify the contribution of individual features to the model predictions.

The implementation of multiple machine learning algorithms enabled a comprehensive comparison of different classification approaches and helped identify the most suitable model for stroke risk prediction.

Table 5.4 Machine Learning Models Implemented

Models	Learning Type	Purpose
Logistic Regression	Linear Classifier	Baseline model for binary classification
Random Forest	Ensemble Learning	Improve prediction through multiple decision trees
XGBoost	Gradient Boosting	Enhance classification performance
CatBoost	Gradient Boosting	Efficient handling of categorical data
Support Vector Machine (SVM)	Kernel-Based Learning	Classification using optimal hyperplane
K-Nearest Neighbors (KNN)	Instance-Based Learning	Classification based on nearest neighbors

5.7. Hyperparameter tuning

Hyperparameter tuning was performed to optimize the performance of the selected machine learning model. In this study, **GridSearchCV** was applied to the **CatBoost** classifier to determine suitable values for important hyperparameters, including **depth**, **learning rate**, and **number of iterations**. The tuning process aimed to improve the model's predictive capability while reducing the risk of overfitting.

The optimized model achieved a high cross-validation score during training. However, when evaluated on the unseen test dataset, it exhibited a decrease in recall and ROC-AUC score compared to the original CatBoost model. Since healthcare applications require reliable identification of stroke cases, the original CatBoost model demonstrated better.

CHAPTER 6. EXPERIMENTAL RESULTS AND DISCUSSION

6.1. Introduction

The performance of the proposed stroke risk prediction system was evaluated using six supervised machine learning algorithms: Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). All models were trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset and evaluated using the same testing dataset to ensure a fair comparison.

The experimental evaluation was carried out using multiple performance metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC score, and Confusion Matrix. In addition, SHapley Additive exPlanations (SHAP) was employed to interpret the prediction outcomes by identifying the contribution of individual healthcare features to each model's decisions. The results obtained from each model are presented and discussed in the following sections.

6.2. Logistic regression

Logistic Regression was implemented as the baseline machine learning model for stroke risk prediction. The model was trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset. Its performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion Matrix, and SHAP explainability analysis.

```
print(confusion_matrix(y_test, y_pred))  
[[728 232]  
 [ 15  47]]
```

Figure 6.1: Confusion Matrix of Logistic Regression

```
[47]: y_prob = lr_model.predict_proba(X_test)[:,-1]  
print("ROC-AUC Score:", roc_auc_score(y_test, y_prob))  
ROC-AUC Score: 0.8494791666666667
```

Figure 6.2: ROC-AUC Score of Logistic Regression

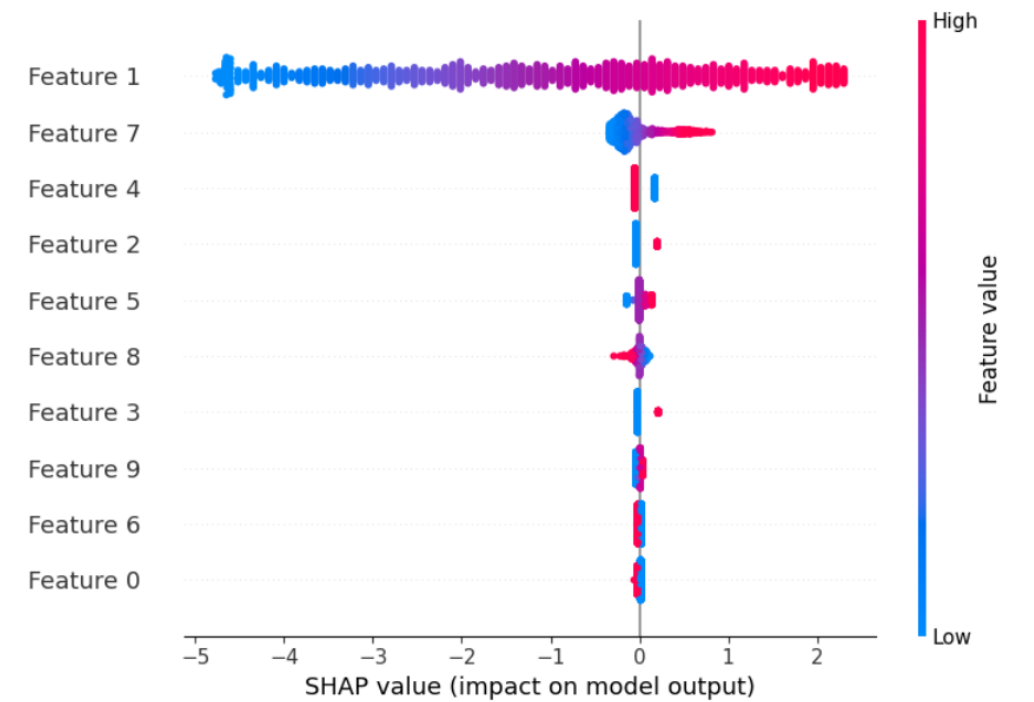


Figure 6.3: SHAP Summary Plot of Logistic Regression

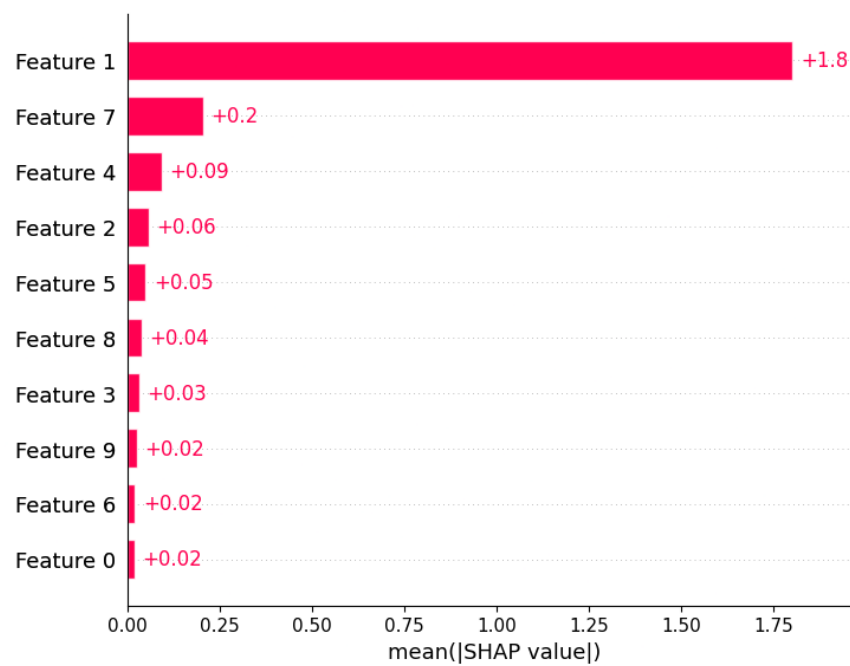


Figure 6.4: SHAP Bar Plot of Logistic Regression

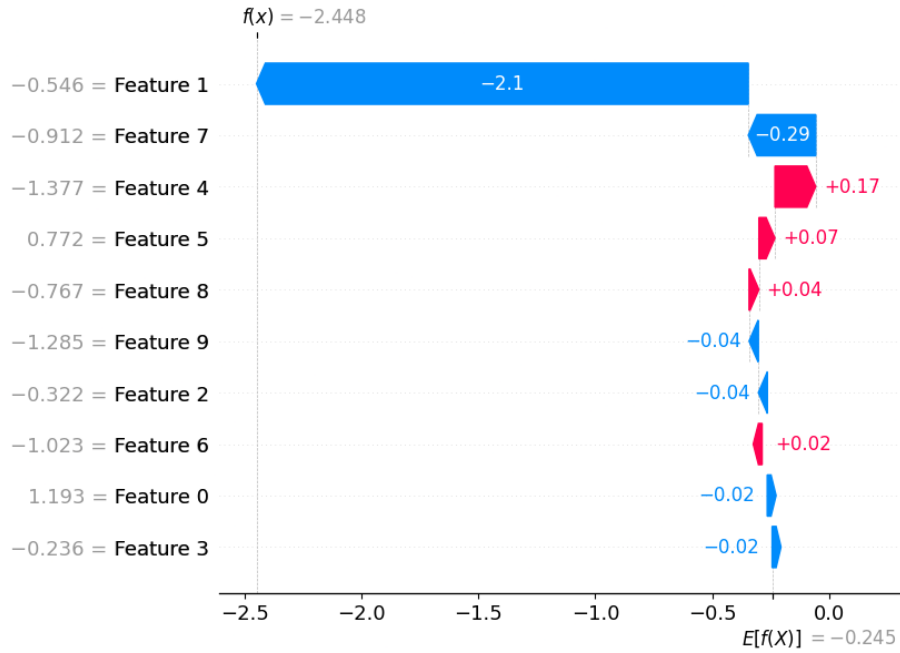


Figure 6.5: SHAP Waterfall Plot of Logistic Regression

Observations

The Logistic Regression model achieved an **accuracy of 75.83%** and a **ROC-AUC score of 84.94%**. Although its overall accuracy was lower than that of the ensemble models, it obtained the **highest recall (76%)** for the stroke class, demonstrating its effectiveness in identifying patients at risk of stroke. The SHAP analysis indicated that **age, BMI, average glucose level, smoking status, hypertension, and heart disease** were the most influential features contributing to the prediction outcomes. These results suggest that Logistic Regression provides strong stroke detection capability while maintaining good model interpretability.

6.3. Random forest

The Random Forest model was implemented to evaluate the performance of an ensemble learning approach for stroke risk prediction. The model was trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset. Its performance was assessed using Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion Matrix, and SHAP explainability analysis.

```
[[932 28]
 [ 55  7]]
ROC-AUC Score: 0.8059391801075269
```

Figure 6.6: Confusion Matrix and ROC-AUC score of Random Forest

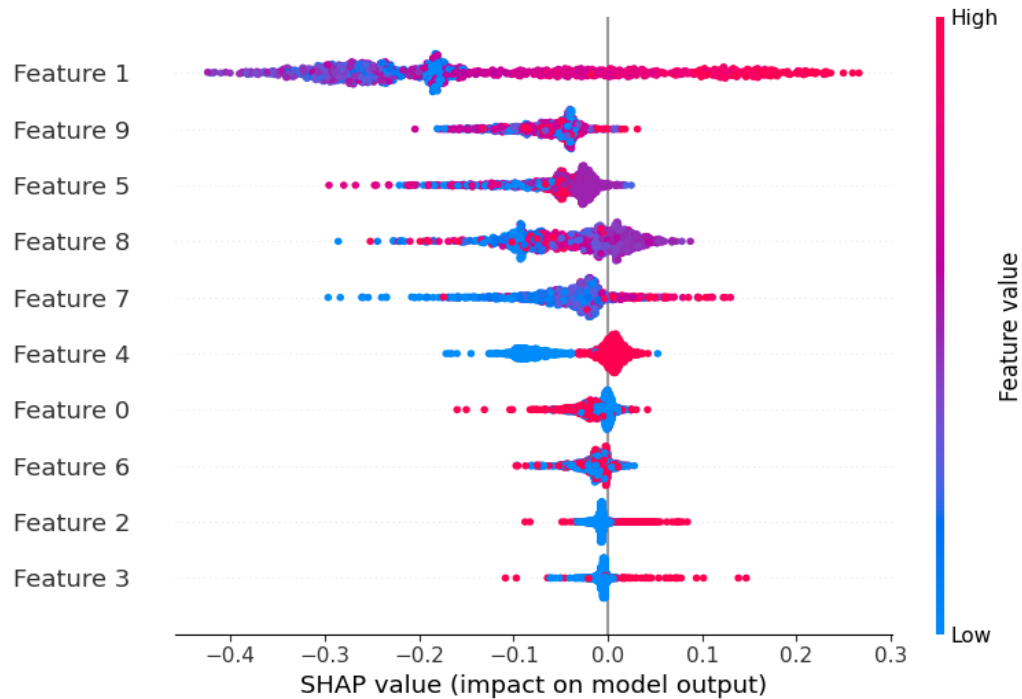


Figure 6.7: SHAP Summary Plot of Random Forest

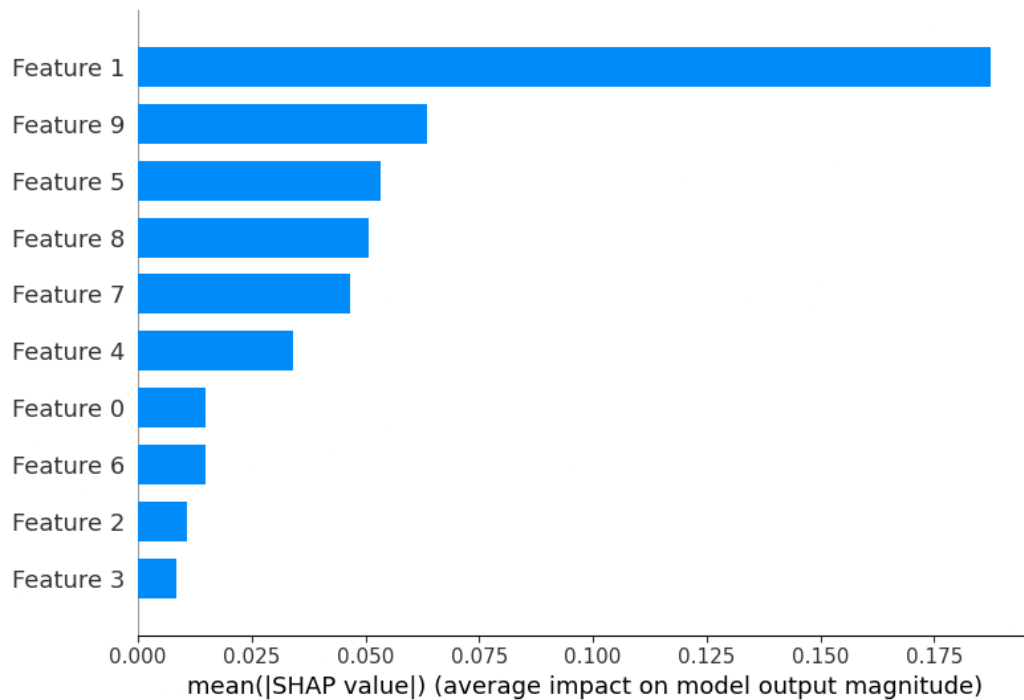


Figure 6.8: SHAP Bar Plot of Random Forest

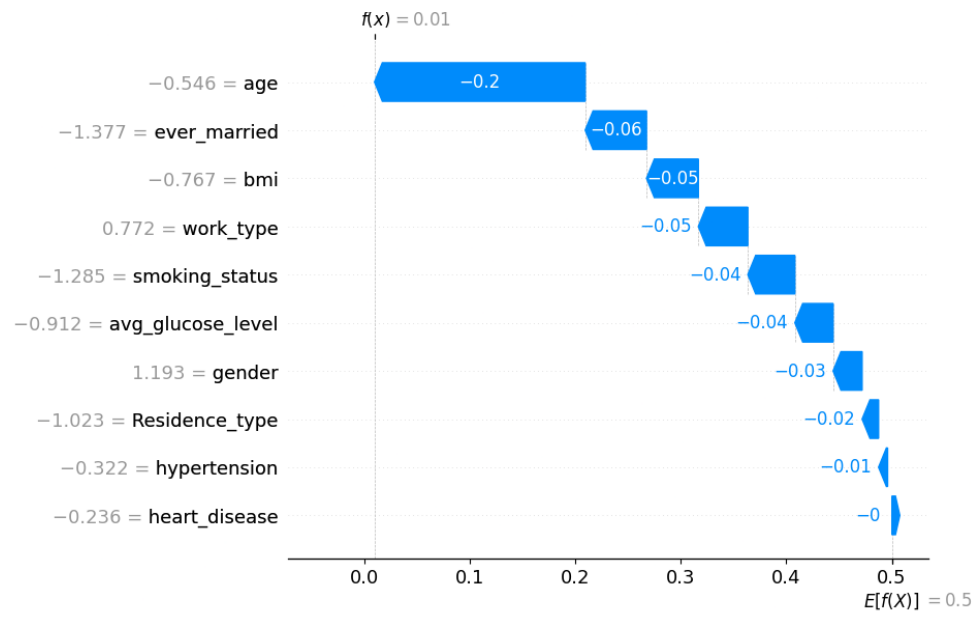


Figure 6.9: SHAP Waterfall Plot of Random Forest

Observations

The Random Forest model achieved an **accuracy of 91.88%**, which was the highest among all the implemented machine learning models. It also obtained a **ROC-AUC score of 80.59%**, demonstrating good classification capability. However, the model achieved a **recall of only 11%** for the stroke class, indicating that it failed to identify many actual stroke cases. The SHAP analysis identified **age, smoking status, work type, BMI, average glucose level, and marital status** as the most influential features affecting prediction outcomes. Although the model demonstrated excellent overall accuracy, its low recall suggests that accuracy alone is not sufficient for evaluating healthcare prediction models, where identifying actual stroke cases is of greater importance.

6.4. XGBoost

The XGBoost model was implemented to evaluate the performance of a gradient boosting algorithm for stroke risk prediction. The model was trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset. Its performance was assessed using Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion Matrix, and SHAP explainability analysis.

```
[[898  62]
 [ 53   9]]
ROC-AUC Score: 0.796606182795699
```

Figure 6.10: Confusion Matrix and ROC-AUC score of XGBoost

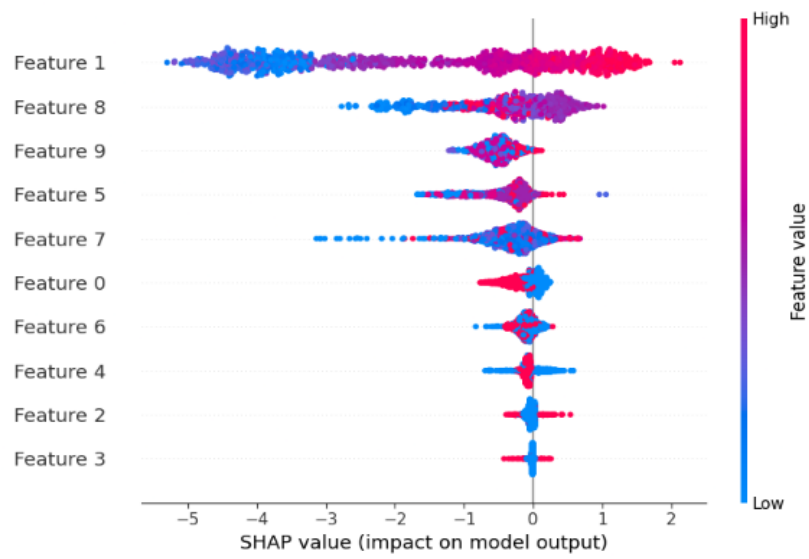


Figure 6.11: SHAP Summary Plot of XGBoost

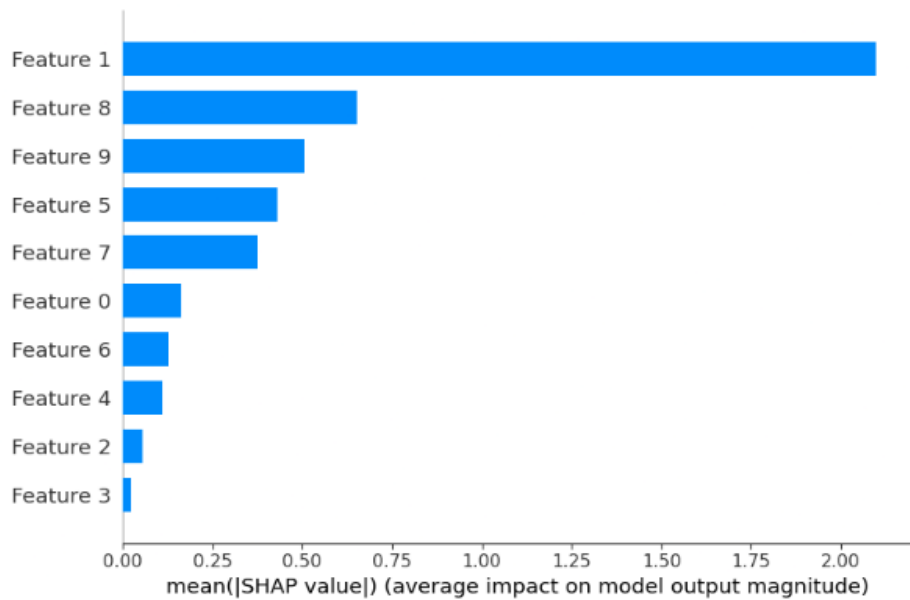


Figure 6.12: SHAP Bar Plot of XGBoost

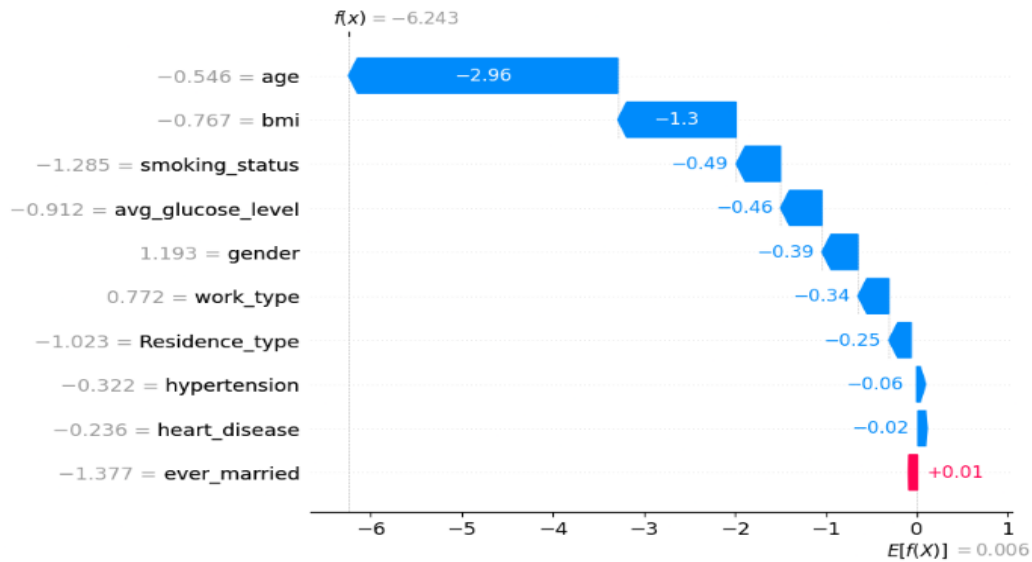


Figure 6.13: SHAP Waterfall Plot of XGBoost

Observations

The XGBoost model achieved an **accuracy of 88.75%** and a **ROC-AUC score of 79.66%**, indicating good overall classification performance. However, the model achieved a **recall of only 15%** for the stroke class, showing limited ability to correctly identify stroke patients. SHAP analysis identified **age, BMI, smoking status, average glucose level, work type, and residence type** as the most influential features affecting prediction outcomes. Although XGBoost achieved high overall accuracy, its lower recall indicates that it was less effective in detecting minority stroke cases.

6.5. CatBoost

The CatBoost model was implemented to evaluate the effectiveness of a gradient boosting algorithm for stroke risk prediction. The model was trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset. Its performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion Matrix, and SHAP explainability analysis.

```
[[856 104]
 [ 37  25]]
ROC-AUC Score: 0.8276041666666668
```

Figure 6.14: Confusion Matrix of CatBoost

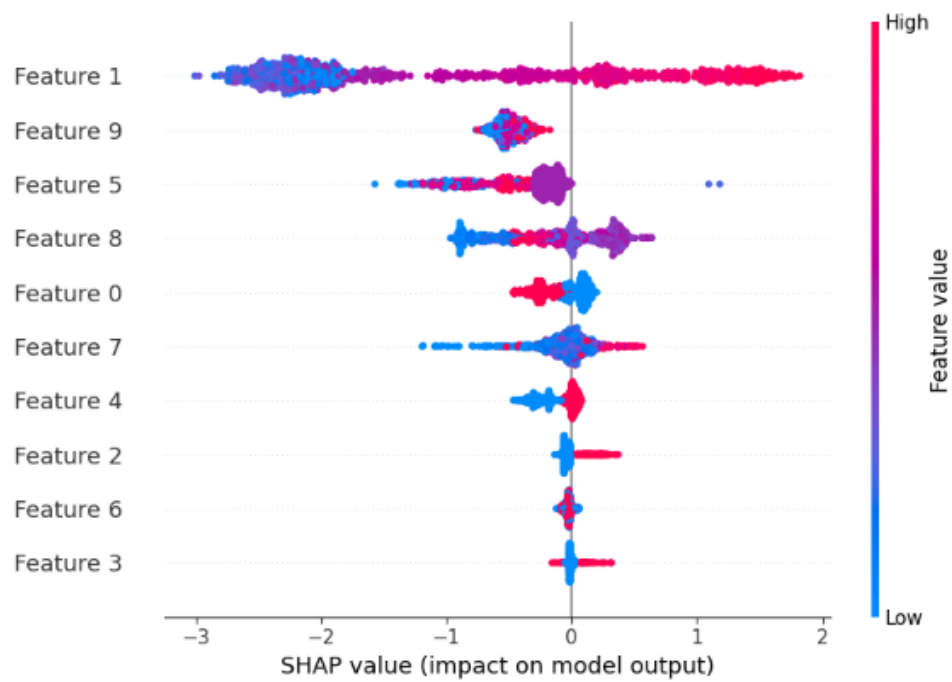


Figure 6.15: SHAP Summary Plot of CatBoost

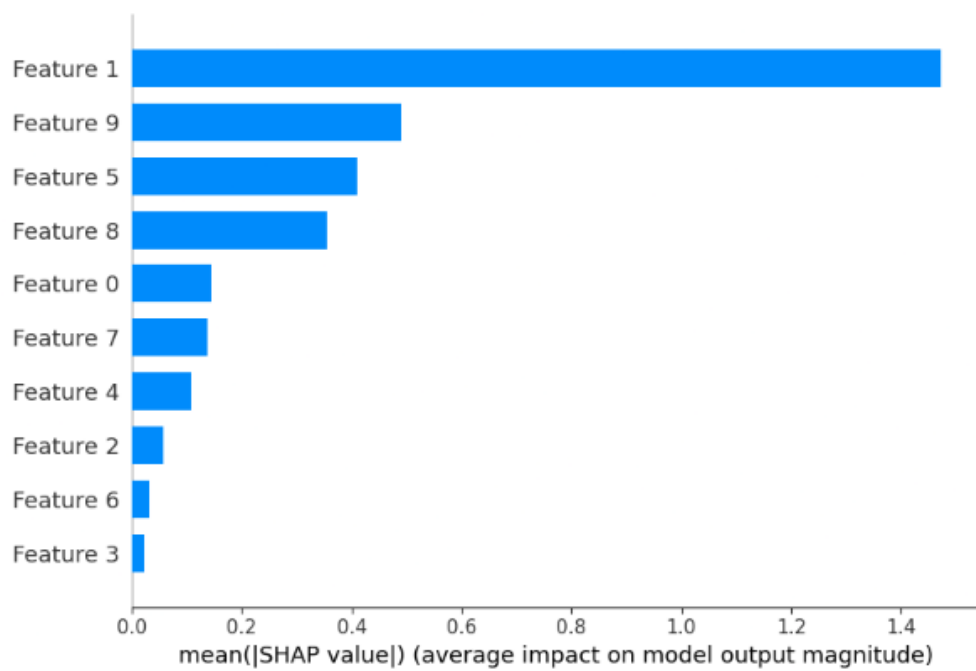


Figure 6.16: SHAP Bar Plot of CatBoost

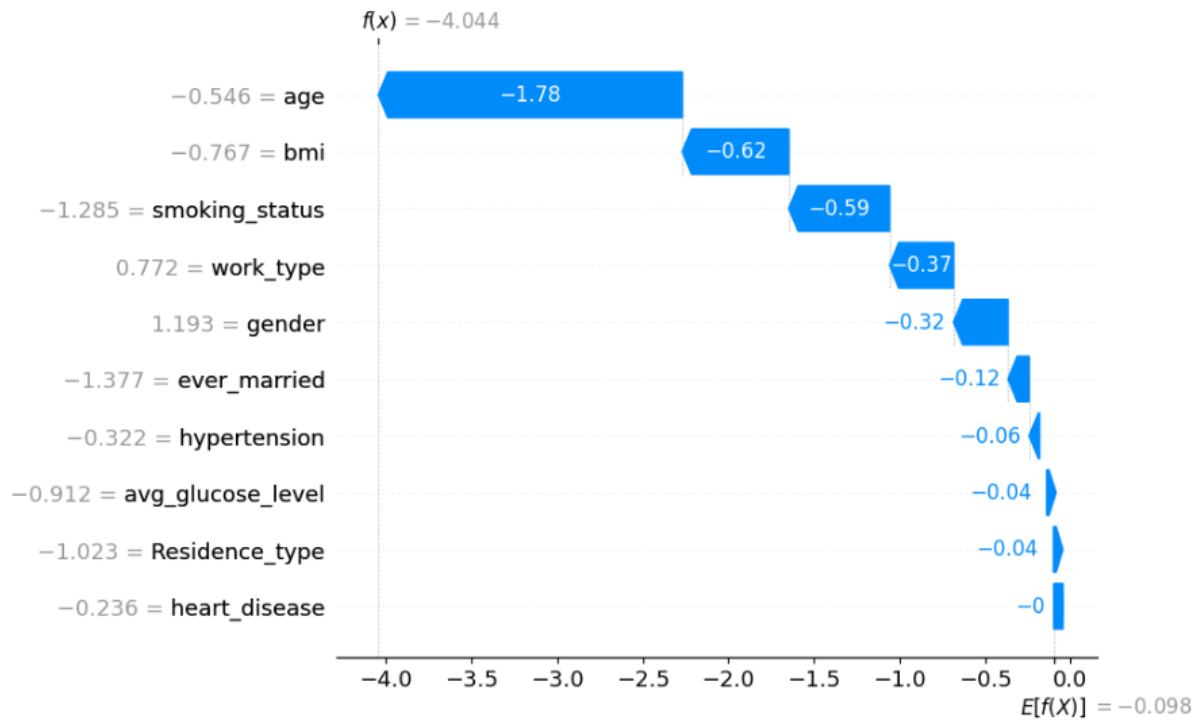


Figure 6.17: SHAP Waterfall Plot of CatBoost

Observations

The CatBoost model achieved an **accuracy of 86.20%** and a **ROC-AUC score of 82.76%**, demonstrating strong classification performance for stroke prediction. The model also achieved a **recall of 40%** for the stroke class, indicating better minority class detection than Random Forest and XGBoost. SHAP analysis identified **age, BMI, smoking status, work type, average glucose level, hypertension, and heart disease** as the most influential features affecting prediction outcomes. The results indicate that CatBoost provided consistent performance while maintaining good interpretability through SHAP visualizations.

6.6. Support vector machine(SVM)

The Support Vector Machine (SVM) model was implemented to evaluate the performance of a kernel-based classification algorithm for stroke risk prediction. The model was trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset. Its performance was assessed using Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion Matrix, and SHAP explainability analysis.

```
[[761 199]
 [ 29  33]]
ROC-AUC Score: 0.766633064516129
```

Figure 6.18: Confusion Matrix and ROC-AUC score of SVM

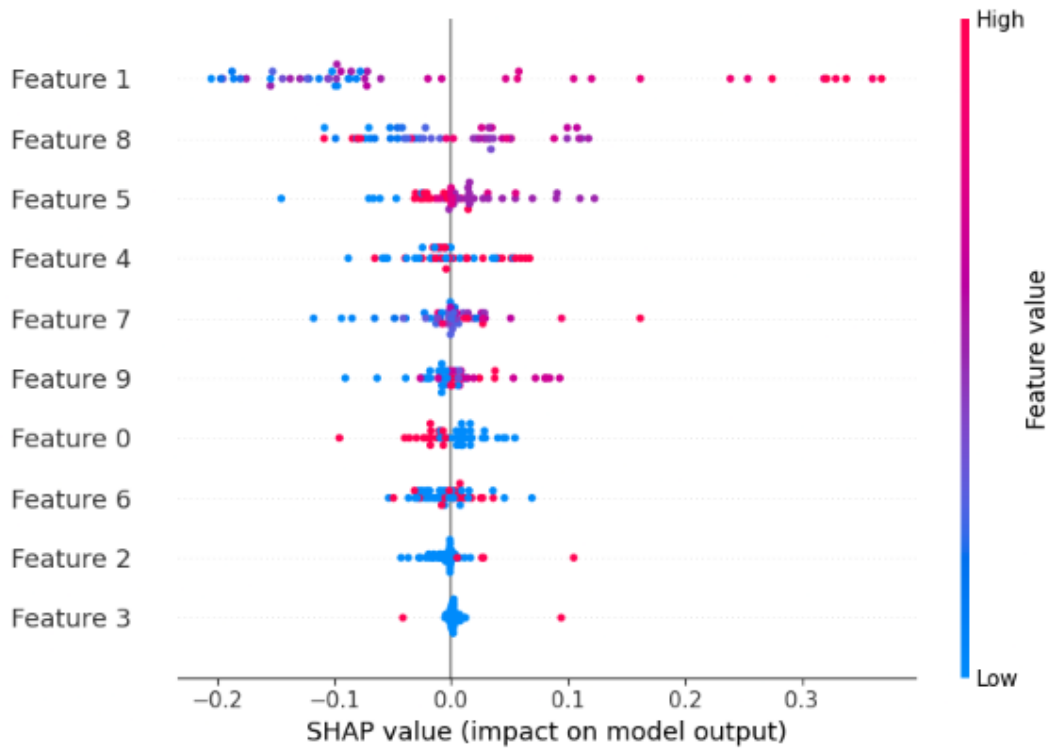


Figure 6.19: SHAP Summary Plot of SVM

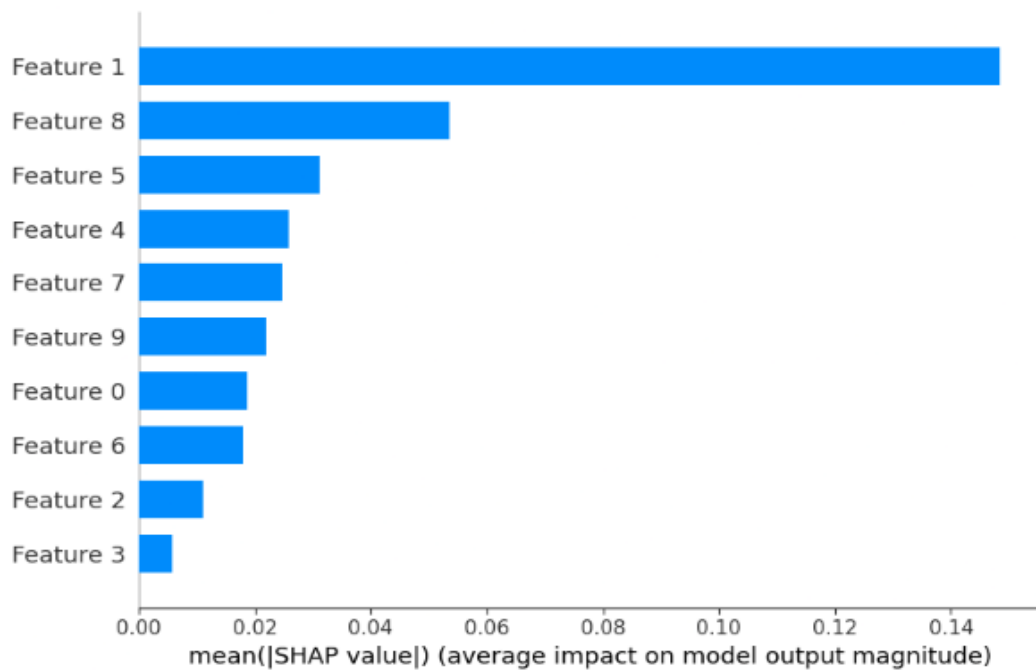


Figure 6.20: SHAP Bar Plot of SVM

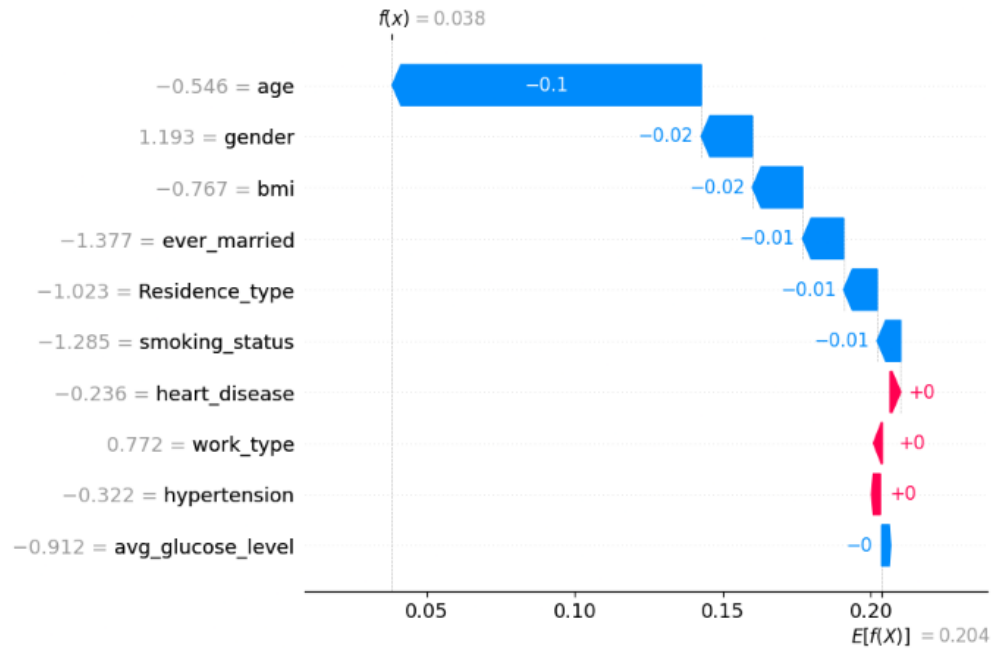


Figure 6.21: SHAP Waterfall Plot of SVM

Observations

The SVM model achieved an **accuracy of 77.69%** and a **ROC-AUC score of 76.66%** on the testing dataset. The model obtained a **recall of 53%** for the stroke class, demonstrating better minority class detection than Random Forest and XGBoost. SHAP analysis identified **age, BMI, smoking status, marital status, residence type, and work type** as the most influential features contributing to stroke prediction. Overall, the SVM model provided moderate classification performance while maintaining good interpretability through SHAP analysis.

6.7. K-Nearest Neighbor (KNN)

The K-Nearest Neighbors (KNN) model was implemented to evaluate the performance of an instance-based learning algorithm for stroke risk prediction. The model was trained using the preprocessed and SMOTE-balanced Healthcare Stroke Prediction Dataset. Its performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC score, Confusion Matrix, and SHAP explainability analysis.

```
[[809 151]
 [ 34  28]]
ROC-AUC Score: 0.6811239919354839
```

Figure 6.22: Confusion Matrix and ROC-AUC score of KNN

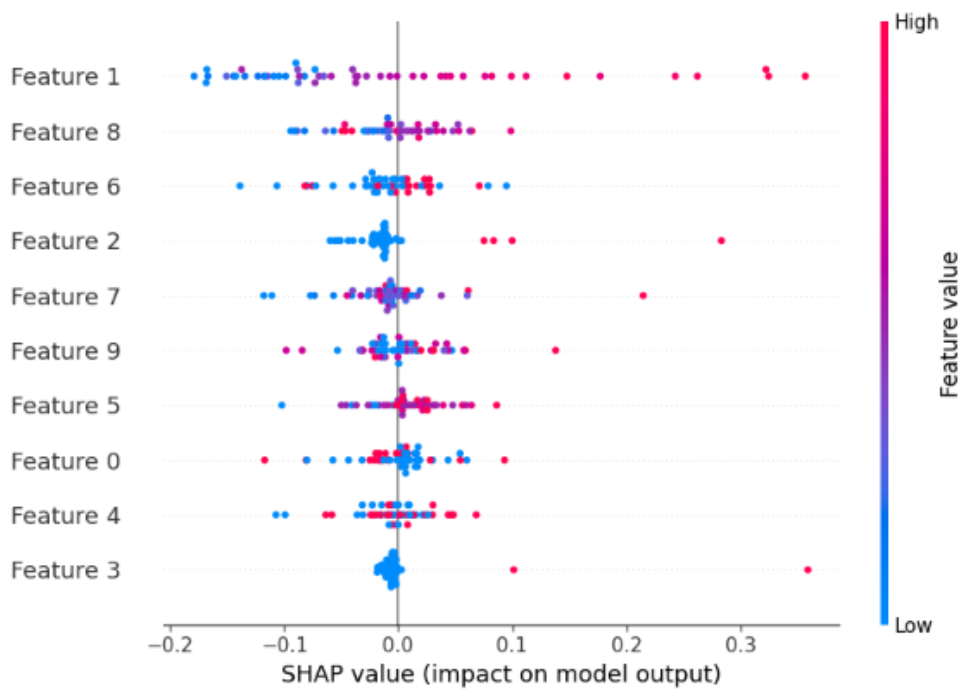


Figure 6.23: SHAP Summary Plot of KNN

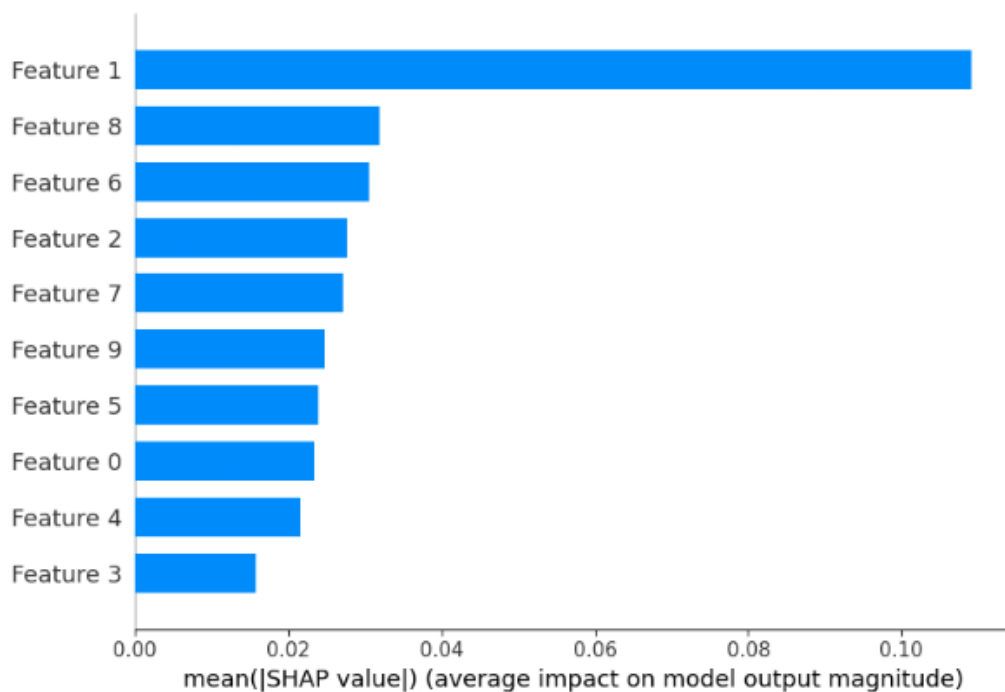


Figure 6.24: SHAP Bar Plot of KNN

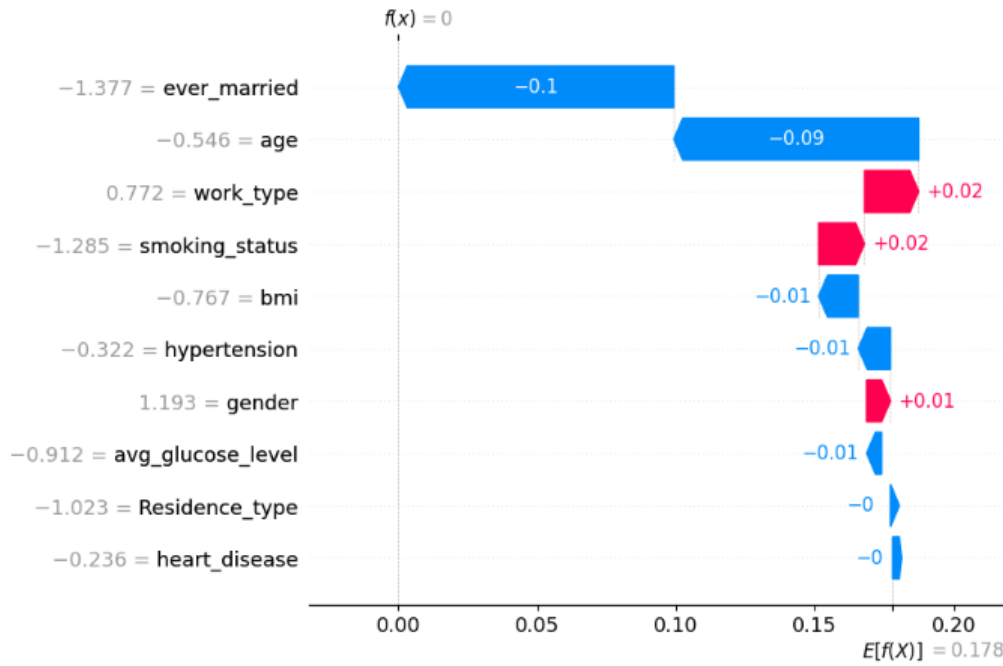


Figure 6.25: SHAP Waterfall Plot of KNN

Observations

The K-Nearest Neighbors (KNN) model achieved an **accuracy of 81.89%** and a **ROC-AUC score of 68.11%** on the testing dataset. The model obtained a **recall of 45%** for the stroke class, indicating moderate performance in identifying stroke cases. SHAP analysis revealed that **age, BMI, smoking status, hypertension, average glucose level, and work type** were the most influential features affecting the prediction outcomes. Although KNN achieved acceptable classification accuracy, its comparatively lower ROC-AUC score indicates limitations in distinguishing between stroke and non-stroke cases

6.8. Comparative performance analysis

To evaluate the effectiveness of the proposed stroke risk prediction system, the performance of all six machine learning models was compared using multiple evaluation metrics, including Accuracy, Precision, Recall, F1-score, and ROC-AUC score. Since stroke prediction involves an imbalanced dataset, multiple performance metrics were considered to provide a comprehensive assessment rather than relying solely on accuracy. The comparative results are presented in **Table 6.2**.

Table 6.2 Comparative Performance of Machine Learning Models

Parameter	Logistic Regressi on	Random Forest	XGBoost	CatBoost	SVM	KNN
Algorithm Type	Linear	Ensemble (Bagging)	Ensemble (Boosting)	Gradient Boosting	Kernel- Based	Instance -Based
Accuracy	75.83%	91.88%	88.75%	86.20%	77.69%	81.89%
Precision (Stroke)	17%	20%	13%	19%	14%	16%
Recall (Stroke)	76%	11%	15%	40%	53%	45%
F1-Score (Stroke)	28%	14%	14%	26%	22%	23%
ROC-AUC Score	84.94%	80.59%	79.66%	82.76%	76.66%	68.11%
SHAP Explainability	✓	✓	✓	✓	✓	✓
Main Strength	Highest Recall & ROC- AUC	Highest Accuracy	Good Accuracy	Balanced Performance	Good Recall	Simple & Interpre table
Main Limitation	Lower Accuracy	Very Low Recall	Low Recall	Slightly Lower Accuracy	Moderate Accuracy	Lowest ROC- AUC

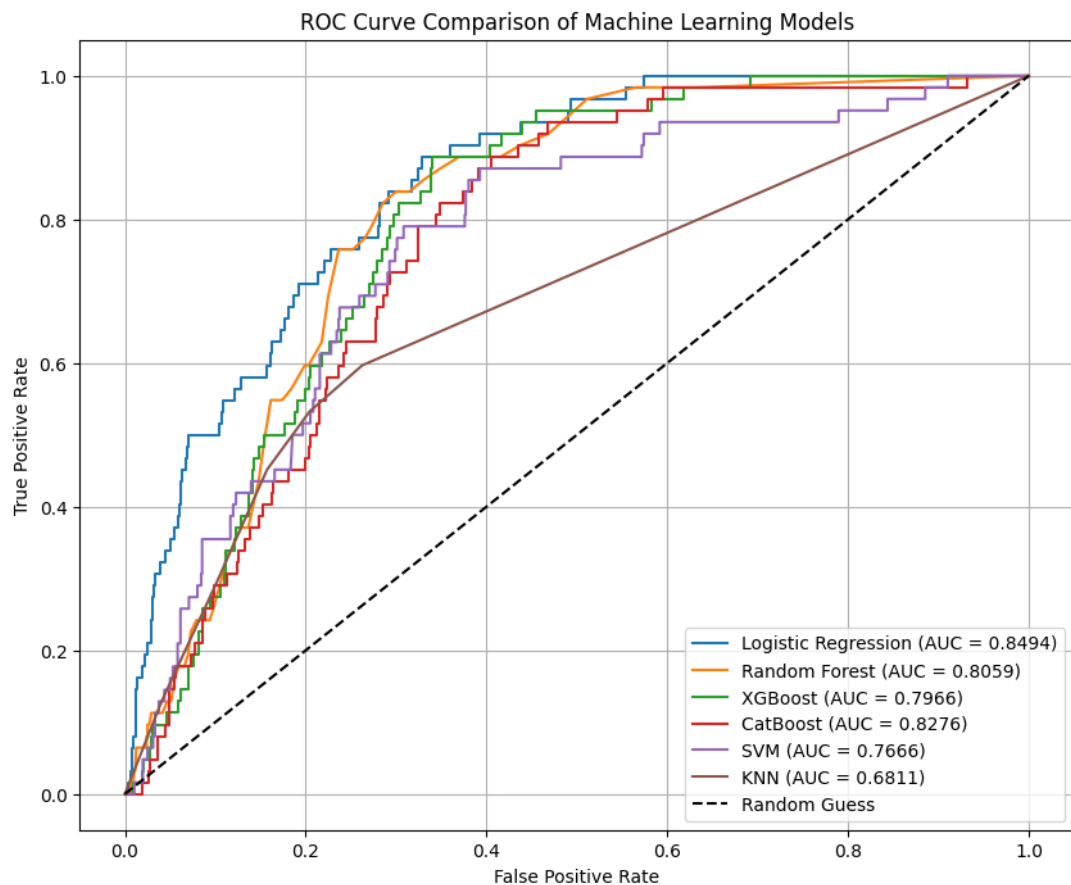


Figure 6.26: ROC curve comparison of machine learning models

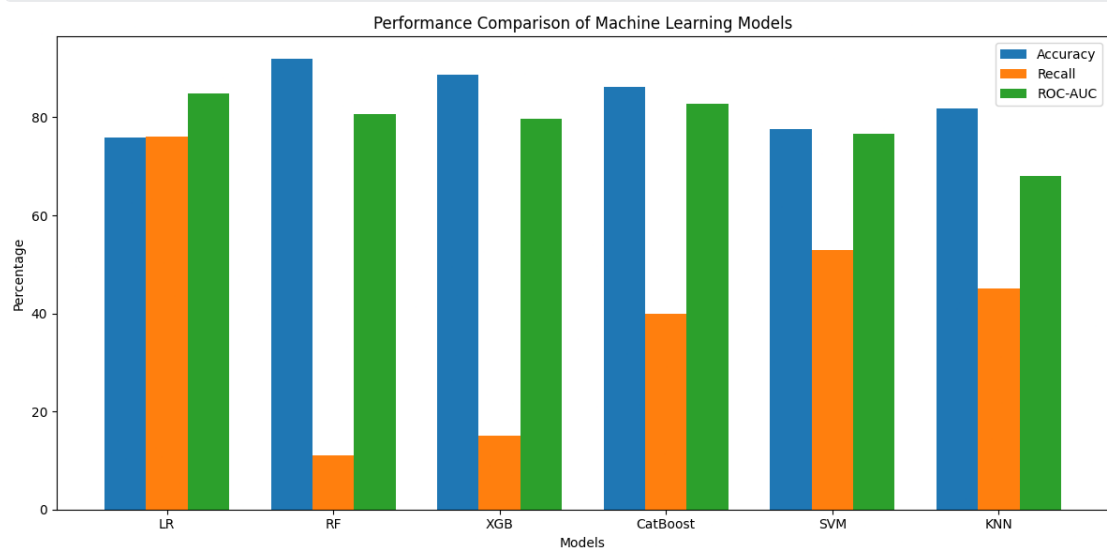


Figure 6.27: Performance comparison of machine learning models

The comparative evaluation demonstrates that each machine learning model exhibited different strengths and limitations. Random Forest achieved the highest overall accuracy (91.88%), indicating excellent performance in correctly classifying the majority of test

samples. However, its low recall (11%) suggests that it failed to identify a significant number of actual stroke cases, making it less suitable for healthcare applications where detecting positive cases is critical.

Logistic Regression achieved the highest recall (76%) and the highest ROC-AUC score (84.94%), demonstrating its effectiveness in identifying stroke cases and distinguishing between the two classes. Although its overall accuracy was lower than the ensemble models, its ability to detect positive cases makes it a reliable baseline model for stroke prediction.

Among the ensemble learning approaches, CatBoost demonstrated a balanced performance by achieving high accuracy (86.20%), a recall of 40%, and a ROC-AUC score of 82.76%. This balance between predictive performance and minority class detection makes CatBoost a suitable choice for healthcare prediction tasks.

The SVM and KNN models achieved moderate performance, while XGBoost provided high accuracy but relatively low recall for the stroke class. These results highlight that accuracy alone is not sufficient for evaluating healthcare prediction systems. Metrics such as recall and ROC-AUC are equally important because they measure the model's ability to correctly identify patients at risk of stroke.

Overall, the comparative analysis indicates that CatBoost provided the most balanced performance across the evaluation metrics, while Logistic Regression demonstrated the strongest capability for detecting stroke cases. These findings support the selection of CatBoost as the final model for the proposed explainable stroke risk prediction system.

6.9. SHAP explainability analysis

Explainability is an important requirement in healthcare applications, as clinicians need to understand the factors influencing a model's predictions before relying on its recommendations. In this study, SHapley Additive exPlanations (SHAP) was employed to interpret the predictions generated by the implemented machine learning models. SHAP

assigns an importance value to each feature, allowing the contribution of individual variables to be visualized and understood.

The SHAP analysis consistently identified age, Body Mass Index (BMI), average glucose level, smoking status, hypertension, and heart disease as the most influential features contributing to stroke risk prediction across the evaluated models. These findings are consistent with established clinical knowledge, indicating that both medical conditions and lifestyle-related factors play a significant role in determining stroke risk.

The generated SHAP visualizations—including the Summary Plot, Bar Plot, and Waterfall Plot—provided both global and local explanations of the prediction outcomes. The Summary Plot illustrated the overall impact of each feature on the model predictions, the Bar Plot ranked the features according to their importance, and the Waterfall Plot explained how individual features contributed to a specific prediction. These visualizations enhanced the transparency of the proposed system and increased confidence in the model's decision-making process.

Overall, the integration of SHAP improved the interpretability of the machine learning models by providing meaningful explanations for prediction outcomes. This makes the proposed stroke risk prediction system more suitable for healthcare applications, where understanding the reasoning behind predictions is as important as achieving high predictive performance.

(The SHAP Summary, Bar, and Waterfall plots presented in Sections 6.2–6.7 demonstrate the contribution of individual features to the prediction outcomes of each machine learning model.)

6.10. Result Discussion

The experimental evaluation demonstrates that the performance of machine learning models for stroke risk prediction varies depending on the learning algorithm and evaluation metric considered. Although **Random Forest** achieved the highest overall accuracy, its low recall indicates that it was less effective in identifying actual stroke cases. In contrast,

Logistic Regression demonstrated the highest recall and ROC-AUC score, highlighting its effectiveness in detecting patients at risk of stroke.

Among the evaluated models, **CatBoost** provided the most balanced performance by achieving competitive accuracy while maintaining satisfactory recall and ROC-AUC scores. Furthermore, the integration of **SHapley Additive exPlanations (SHAP)** improved the transparency of the prediction process by identifying the healthcare features that contributed most significantly to stroke risk prediction. This enhances the reliability and interpretability of the proposed system for healthcare applications.

Overall, the experimental results indicate that combining machine learning techniques with explainable artificial intelligence can provide accurate and interpretable stroke risk prediction. The findings also emphasize that evaluation metrics such as **recall** and **ROC-AUC** are essential for healthcare applications, where correctly identifying high-risk patients is more important than achieving high accuracy alone.

CHAPTER 7. PROPOSED DEPLOYMENT FRAMEWORK

7.1. Proposed streamlit system

The proposed deployment of the stroke risk prediction system is based on the **Streamlit** framework, which provides a simple and interactive web-based interface for machine learning applications. Although the deployment was not implemented as part of this project, Streamlit is proposed as a suitable platform due to its ease of integration with Python-based machine learning models and its ability to provide real-time predictions through a user-friendly interface.

In the proposed system, healthcare professionals would enter patient information such as age, gender, hypertension status, heart disease, body mass index (BMI), average glucose level, smoking status, work type, marital status, and residence type through a web interface. The trained machine learning model would process the input data and generate a stroke risk prediction. To improve transparency, SHAP explanations could also be displayed to highlight the factors that most influenced the prediction, allowing clinicians to better understand the model's decision-making process.

The proposed Streamlit application is intended to support healthcare professionals by providing a fast, interpretable, and accessible decision-support tool for stroke risk assessment. The conceptual design of the proposed system is presented in the following sections.

7.2. User interface design

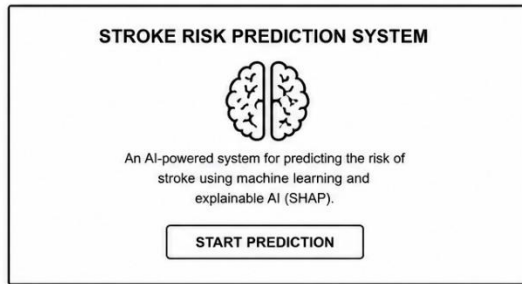


Figure 7.1: Home Screen

Figure 7.2: Patient Information Form

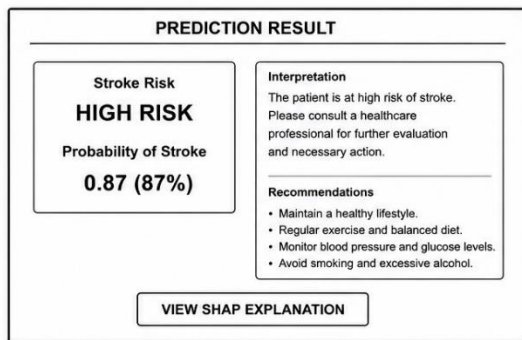


Figure 7.3: Prediction Result Screen

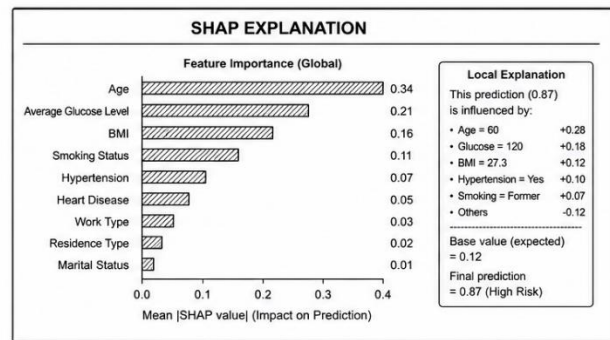


Figure 7.4: SHAP Explanation Screen

(Note: The above interfaces are design prototypes of the proposed system and not actual implementation screenshots.)

7.3. Deployment architecture

The proposed stroke risk prediction system is designed to be deployed as a web-based application using the **Streamlit** framework. In the proposed architecture, the trained machine learning model, preprocessing components, and SHAP explainability module are integrated into a single application to provide real-time stroke risk prediction.

When a healthcare professional enters patient information through the Streamlit interface, the input data is first processed using the same preprocessing pipeline employed during model development. This includes applying the saved preprocessing techniques before passing the processed data to the trained machine learning model. The model then predicts the patient's stroke risk and generates the corresponding probability score. To improve transparency, the SHAP module provides an explanation of the prediction by identifying the features that contributed most significantly to the result.

The proposed deployment architecture enables healthcare professionals to obtain accurate and interpretable predictions through a simple web interface, making the system suitable for clinical decision support and future healthcare applications.

7.4. Prediction workflow

The proposed stroke risk prediction system follows a structured workflow to generate predictions based on patient information. Initially, the healthcare professional enters the required patient details, including demographic, medical, and lifestyle-related attributes, through the Streamlit web interface. The entered information is then validated and preprocessed using the same techniques applied during model development.

After preprocessing, the processed data is passed to the trained machine learning model, which predicts the probability of stroke occurrence. The prediction result is displayed to the user along with the corresponding stroke risk category. To improve transparency and interpretability, the SHAP module provides an explanation of the prediction by highlighting the features that contributed most significantly to the final decision. This workflow enables healthcare professionals to understand both the prediction and the reasoning behind it, supporting informed clinical decision-making.

7.5. Advantages

The proposed Streamlit-based stroke risk prediction system offers several advantages for healthcare professionals by combining machine learning with explainable artificial intelligence. Some of the key advantages are as follows:

- Provides early stroke risk prediction, enabling timely medical intervention.
- Delivers fast and automated predictions using machine learning algorithms.
- Enhances transparency through SHAP-based explanations of model predictions.
- Supports healthcare professionals in clinical decision-making.
- Offers a simple and user-friendly web interface for entering patient information.
- Can be easily deployed on cloud platforms for remote accessibility.
- Reduces manual effort in preliminary stroke risk assessment.
- Can be updated and retrained with new healthcare data to improve prediction performance over time.

7.6. Challenges during real-world deployment

Although the proposed stroke risk prediction system demonstrates promising performance, several challenges may arise during real-world deployment in healthcare environments. Ensuring the privacy and security of patient data is a major concern, as healthcare information is highly sensitive and must comply with applicable data protection regulations. In addition, the prediction model may require periodic retraining using updated clinical data to maintain its accuracy and reliability over time.

Another challenge is the integration of the system with existing hospital information systems and Electronic Health Record (EHR) platforms, which may require additional technical development. Furthermore, machine learning predictions should be considered as decision-support tools rather than replacements for clinical judgment, as final medical decisions should always be made by qualified healthcare professionals. Addressing these challenges is essential for the successful adoption of the proposed system in real-world healthcare settings.

7.7. Future enhancements

The proposed stroke risk prediction system can be further enhanced to improve its functionality, usability, and applicability in real-world healthcare environments. Future work may include integrating the system with **Electronic Health Record (EHR)** systems to enable automatic retrieval of patient information and support real-time clinical decision-making. The prediction performance can also be improved by training the models on larger and more diverse healthcare datasets collected from multiple hospitals and regions.

Future enhancements may also include the implementation of **deep learning models** and the evaluation of additional explainable AI techniques to improve both prediction accuracy and interpretability. Furthermore, the proposed Streamlit application can be deployed on a secure cloud platform, allowing healthcare professionals to access the system remotely. Continuous model updates and validation using new clinical data will

further improve the reliability and effectiveness of the proposed stroke risk prediction system.

CHAPTER 8. CONCLUSION AND FUTURE SCOPE

8.1. Conclusion

This project successfully developed and evaluated an explainable machine learning-based system for early stroke risk prediction using a publicly available healthcare dataset. The dataset was preprocessed through data cleaning, categorical feature encoding, feature scaling, and class balancing using the SMOTE technique to improve the quality of the training data. Six machine learning algorithms—Logistic Regression, Random Forest, XGBoost, CatBoost, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)—were implemented and evaluated using accuracy, precision, recall, F1-score, ROC-AUC score, confusion matrix, and SHAP explainability analysis.

The comparative evaluation demonstrated that each algorithm exhibited different strengths. Random Forest achieved the highest classification accuracy, whereas Logistic Regression obtained the highest recall and ROC-AUC score, making it highly effective in identifying stroke cases. Among the advanced machine learning algorithms, CatBoost demonstrated the most balanced overall performance by maintaining good accuracy while achieving better stroke detection capability than the other ensemble models. SHAP explainability analysis further improved the transparency of the prediction system by identifying clinically significant factors such as age, BMI, average glucose level, hypertension, heart disease, and smoking status as the primary contributors to stroke risk.

An additional hyperparameter tuning experiment was performed on the CatBoost model using GridSearchCV. Although the optimized model achieved a high cross-validation score during training, its performance on the unseen test dataset declined, particularly in terms of recall and ROC-AUC. Therefore, the original CatBoost model was retained, highlighting the importance of evaluating machine learning models on independent test data rather than relying solely on training performance.

Overall, the study demonstrates that combining machine learning techniques with Explainable Artificial Intelligence (XAI) can provide accurate, transparent, and clinically meaningful stroke risk prediction. The developed system has the potential to support

healthcare professionals in early risk assessment and decision-making while improving trust through interpretable model predictions.

8.2. Limitations

Despite the promising results, the proposed system has several limitations. The study was conducted using a single publicly available dataset, which may not fully represent the diversity of real-world patient populations. The performance of the developed models may vary when applied to data collected from different hospitals or geographical regions.

In addition, the proposed system has not been deployed or validated in a real clinical environment. The machine learning models should therefore be considered as decision-support tools rather than replacements for professional medical judgment. Further validation using larger clinical datasets is required before practical implementation.

8.3. Future Scope

The proposed stroke prediction system can be further enhanced by incorporating larger and more diverse healthcare datasets collected from multiple hospitals and healthcare organizations. A larger dataset would improve the generalizability and robustness of the developed models across different patient populations.

Future research may explore advanced deep learning architectures, such as Artificial Neural Networks (ANNs), Long Short-Term Memory (LSTM) networks, or Transformer-based models, to capture more complex relationships among patient attributes and potentially improve prediction performance.

Further model optimization can be achieved through advanced hyperparameter tuning techniques, feature engineering, and automated machine learning (AutoML) frameworks. Additional explainability methods such as LIME or counterfactual explanations may also be incorporated to provide complementary insights into individual predictions.

The selected model can be deployed as a real-time web or mobile application using frameworks such as Flask or Streamlit, enabling healthcare professionals to assess stroke risk quickly during routine clinical practice. Integration with Electronic Health Record

(EHR) systems could further automate patient risk assessment and support timely medical intervention.

Finally, future work may include prospective clinical validation using real-world patient data to evaluate the reliability, accuracy, and practical applicability of the proposed prediction system in hospital environments.

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